Compute Cell Scheduling Records

This document records how tasks were scheduled on small compute clusters. The workload and the machine failures are real records from a production compute cell of about 12,500 machines (the Google cluster trace of May 2011, clusterdata-2011-2, CC-BY 4.0): every task's submission time, job, priority, CPU and memory request and run time, and every machine's capacity, failure time and return time, are taken from that trace (ids are renamed). Where and when each task ran in the small cluster is not recorded in the trace: it is computed from those records by the scheduling policy stated below, applied exactly.

The document is divided into chapters titled 'Cell window <NN>'. Each chapter is an independent run of one small cluster over a window of 1200 seconds. Questions refer to a chapter by that title. Times are whole seconds from the start of the window (time 0); the window covers times 0 to 1200.

Each chapter has four sections. 'Machines' lists the cluster's machines with their CPU and memory capacity, the time a machine failed (if it did) and the time it returned (if it did). 'Task log' lists, in time order, every event of every task from time 0 to the window end: submit, start, finish, evict and fail, with the task's job, priority, CPU and memory request, run time, and the machine involved. 'Events' lists machine failures and returns, evictions and restarts, and the longest waits. 'Queue at window end' gives the state after the events at time 1200: the tasks still running (with their remaining run time) and the tasks still waiting, in queue order.

Tasks. A task belongs to a job and has a priority from 0 to 11 (higher is more important). Priorities 9 to 11 are the *production* tier and priorities 0 and 1 the *free* tier. Each task requests an amount of CPU and of memory, which it holds on one machine while it runs, and has a run time: once started it runs for exactly that many seconds and then finishes, unless it is stopped first. The run time is given on every row of the task log, starting with the task's submit row.

Scheduling policy. Every window starts with all machines empty and no task waiting. At every time at which something happens, in this order:

1. **Finishes.** Tasks whose run time is over finish and free their CPU and memory.

2. **Failures.** Machines that fail at this time go down. Each task running on such a machine stops (a *fail* event) and goes back to the queue.

3. **Returns.** Machines that return at this time come back, empty, with their capacity.

4. **Submissions.** Tasks submitted at this time join the queue.

5. **Scheduling round.** The waiting tasks are taken one at a time in *queue order*: higher priority first, then earlier submission time, then lower task id. Each task, in turn:

a. **First fit.** The task starts on the first machine, in machine id order, that is up and whose free CPU and free memory are both at least the task's requests (subject to the reservation rule in c).

b. **Preemption.** If it fits nowhere, no task holds a reservation yet in this round, and it is a production task, it may evict free-tier tasks from one machine. It looks at the up machines in id order. On each it considers the free-tier tasks running there, lowest priority first, then the most recently started first (then the higher task id first), and evicts them one by one in that order until it fits; it takes the first machine where this works, evicting only as many as needed there, and starts on it. If no machine works, nothing is evicted. An evicted task goes back to the queue at once (it may start again later in the same round); it loses its progress and will run its full run time again.

c. **Reservation and backfilling.** The first task in the round that can start neither by a. nor by b. gets a *reservation*: the earliest time at which it could start, computed from the run times of the tasks now running (assuming no new submissions, failures or returns), on the first machine, in id order, where that earliest time is reached. Every later task in the round may still start by first fit, but on the reserved machine only if it will finish no later than the reserved time, or if its requests fit in what the reserved machine will still have free at the reserved time after the reserved task's requests (and after other such starts on it in the same round). On the other machines it starts as usual. Later tasks do not preempt. A task that would not fit on any up machine even if that machine were empty gets no reservation and simply keeps waiting; the next task that cannot start may then get the reservation.

Tasks stopped by an eviction or a failure keep their original submission time (and so their place in queue order) and start again from the beginning. A task never moves while it runs.

Order of rows. Task log rows with the same time follow the steps above: finishes (by task id), failures (by machine, then task id), submissions (by task id), then the scheduling round's starts and evictions in the order the round makes them (each eviction just before the start that caused it).

Terms used in the Events section and in questions. A task's *wait* is the time from its submission to its first start. A task is *restarted* when it starts again after an eviction or a failure. The number of *evictions* counts evict events (one per evicted task). 'Running on a machine at time t' means running after all the events at time t.

The cluster and the workload. Each window has 8 machines, ids M01 to M08. Each is modelled on a real machine of the cell (one of them is a machine that failed during the window in the trace) and has 1/8 of that machine's recorded CPU and memory capacity. A failed machine returns at the time the trace records it coming back, if that is within 6 hours; otherwise it does not return. The tasks are a sample of the real tasks submitted in the cell during the window, chosen by this fixed rule: jobs submitted in the window are taken in a random order, first up to 6 production jobs, then free-tier jobs until the offered load reaches 54%, then other jobs until it reaches 90% (a job that would push it above 95% is skipped). A job contributes its first 4 tasks (16 for a free-tier job) that finished normally in the trace and fit on at least one machine of the window. Offered load = the larger, over CPU and memory, of the sum of request x run time inside the window over all tasks, divided by the cluster's total capacity x the window length.

After the window. For questions about what happens after time 1200: no new task is submitted and no machine fails after the window; a machine that is down returns at its listed return time (if any); the policy continues unchanged from the state given in 'Queue at window end'.

Units. Times and run times in seconds. CPU and memory in units of 1/100,000 of the largest machine in the original cell (the trace's normalised values, rounded to whole units).

Chapters in this document (itops_d03): Cell window 01, Cell window 06, Cell window 09, Cell window 28, Cell window 29.

Cell window 01

Times 0 to 1200. The cluster has 8 machines. 126 tasks of 70 jobs are submitted during the window.

Cell window 01: Machines

machine	CPU	memory	fails at	returns at
M01	6250	9363	-	-
M02	12500	12500	-	-
M03	6250	6244	-	-
M04	6250	3116	359	never
M05	6250	6244	-	-
M06	6250	6244	-	-
M07	6250	3116	-	-
M08	6250	6244	-	-

Cell window 01: Task log

time	task	job	priority	CPU	memory	run time	event	machine
17	T001	J01	6	937	1244	244	submit	-
17	T001	J01	6	937	1244	244	start	M01
38	T002	J02	6	1250	651	221	submit	-
38	T003	J02	6	1250	651	217	submit	-
38	T004	J02	6	1250	651	206	submit	-
38	T005	J02	6	1250	651	215	submit	-
38	T006	J03	6	937	275	77	submit	-
38	T007	J03	6	937	275	74	submit	-
38	T008	J03	6	937	275	74	submit	-
38	T009	J03	6	937	275	98	submit	-
38	T002	J02	6	1250	651	221	start	M01
38	T003	J02	6	1250	651	217	start	M01
38	T004	J02	6	1250	651	206	start	M01
38	T005	J02	6	1250	651	215	start	M01
38	T006	J03	6	937	275	77	start	M02
38	T007	J03	6	937	275	74	start	M02
38	T008	J03	6	937	275	74	start	M02
38	T009	J03	6	937	275	98	start	M02
56	T010	J04	6	625	187	92	submit	-
56	T011	J05	4	62	398	20	submit	-
56	T010	J04	6	625	187	92	start	M02
56	T011	J05	4	62	398	20	start	M01
76	T011	J05	4	62	398	20	finish	M01
76	T012	J06	4	6250	3107	128	submit	-
76	T013	J06	4	6250	3107	116	submit	-
76	T014	J06	4	6250	3107	128	submit	-
76	T015	J06	4	6250	3107	106	submit	-
76	T012	J06	4	6250	3107	128	start	M02
76	T013	J06	4	6250	3107	116	start	M03
76	T014	J06	4	6250	3107	128	start	M04
76	T015	J06	4	6250	3107	106	start	M05
77	T016	J07	4	1250	466	183	submit	-
77	T017	J07	4	1250	466	180	submit	-
77	T018	J07	4	1250	466	181	submit	-
77	T019	J07	4	1250	466	183	submit	-
77	T016	J07	4	1250	466	183	start	M02
77	T017	J07	4	1250	466	180	start	M06
77	T018	J07	4	1250	466	181	start	M06

time	task	job	priority	CPU	memory	run time	event	machine
77	T019	J07	4	1250	466	183	start	M06
112	T007	J03	6	937	275	74	finish	M02
112	T008	J03	6	937	275	74	finish	M02
115	T006	J03	6	937	275	77	finish	M02
119	T020	J08	6	625	187	19	submit	-
119	T020	J08	6	625	187	19	start	M02
136	T009	J03	6	937	275	98	finish	M02
138	T020	J08	6	625	187	19	finish	M02
148	T010	J04	6	625	187	92	finish	M02
168	T021	J09	0	6250	1244	148	submit	-
168	T021	J09	0	6250	1244	148	start	M07
169	T022	J10	0	6250	1244	155	submit	-
169	T022	J10	0	6250	1244	155	start	M08
182	T015	J06	4	6250	3107	106	finish	M05
191	T023	J11	6	625	280	262	submit	-
191	T023	J11	6	625	280	262	start	M02
192	T013	J06	4	6250	3107	116	finish	M03
204	T012	J06	4	6250	3107	128	finish	M02
204	T014	J06	4	6250	3107	128	finish	M04
229	T024	J12	6	937	1244	207	submit	-
229	T024	J12	6	937	1244	207	start	M02
244	T004	J02	6	1250	651	206	finish	M01
245	T025	J13	6	937	278	66	submit	-
245	T026	J13	6	937	278	64	submit	-
245	T027	J13	6	937	278	58	submit	-
245	T028	J13	6	937	278	123	submit	-
245	T025	J13	6	937	278	66	start	M01
245	T026	J13	6	937	278	64	start	M02
245	T027	J13	6	937	278	58	start	M02
245	T028	J13	6	937	278	123	start	M02
253	T005	J02	6	1250	651	215	finish	M01
255	T003	J02	6	1250	651	217	finish	M01
257	T017	J07	4	1250	466	180	finish	M06
258	T018	J07	4	1250	466	181	finish	M06
259	T002	J02	6	1250	651	221	finish	M01
260	T016	J07	4	1250	466	183	finish	M02
260	T019	J07	4	1250	466	183	finish	M06
261	T001	J01	6	937	1244	244	finish	M01
303	T027	J13	6	937	278	58	finish	M02
307	T029	J14	6	937	1244	79	submit	-
307	T029	J14	6	937	1244	79	start	M01
308	T030	J15	6	625	197	78	submit	-
308	T031	J15	6	625	197	72	submit	-
308	T030	J15	6	625	197	78	start	M01
308	T031	J15	6	625	197	72	start	M01
309	T026	J13	6	937	278	64	finish	M02
311	T025	J13	6	937	278	66	finish	M01
316	T021	J09	0	6250	1244	148	finish	M07
324	T022	J10	0	6250	1244	155	finish	M08
327	T032	J16	6	625	197	88	submit	-
327	T033	J16	6	625	197	86	submit	-
327	T032	J16	6	625	197	88	start	M01
327	T033	J16	6	625	197	86	start	M01
337	T034	J17	9	6250	1590	72	submit	-
337	T034	J17	9	6250	1590	72	start	M02
348	T035	J18	0	3125	311	354	submit	-
348	T035	J18	0	3125	311	354	start	M02

time	task	job	priority	CPU	memory	run time	event	machine
368	T028	J13	6	937	278	123	finish	M02
380	T031	J15	6	625	197	72	finish	M01
386	T029	J14	6	937	1244	79	finish	M01
386	T030	J15	6	625	197	78	finish	M01
397	T036	J19	9	6250	233	124	submit	-
397	T036	J19	9	6250	233	124	start	M03
407	T037	J20	0	6250	466	43	submit	-
407	T037	J20	0	6250	466	43	start	M05
409	T034	J17	9	6250	1590	72	finish	M02
411	T038	J21	9	6250	3180	230	submit	-
411	T038	J21	9	6250	3180	230	start	M02
413	T033	J16	6	625	197	86	finish	M01
413	T039	J22	1	1250	1590	353	submit	-
413	T039	J22	1	1250	1590	353	start	M01
415	T032	J16	6	625	197	88	finish	M01
419	T040	J23	6	625	3180	22	submit	-
419	T040	J23	6	625	3180	22	start	M01
436	T024	J12	6	937	1244	207	finish	M02
441	T040	J23	6	625	3180	22	finish	M01
450	T037	J20	0	6250	466	43	finish	M05
453	T023	J11	6	625	280	262	finish	M02
461	T041	J24	8	6250	3107	179	submit	-
461	T041	J24	8	6250	3107	179	start	M05
478	T042	J25	6	625	280	132	submit	-
478	T043	J26	8	6250	4773	1532	submit	-
478	T043	J26	8	6250	4773	1532	start	M06
478	T042	J25	6	625	280	132	start	M01
497	T044	J27	4	2499	3339	137	submit	-
497	T045	J27	4	2499	3339	264	submit	-
497	T046	J27	4	2499	3339	135	submit	-
497	T047	J27	4	2499	3339	138	submit	-
497	T044	J27	4	2499	3339	137	start	M01
497	T045	J27	4	2499	3339	264	start	M02
497	T046	J27	4	2499	3339	135	start	M08
521	T036	J19	9	6250	233	124	finish	M03
521	T047	J27	4	2499	3339	138	start	M03
528	T048	J28	6	937	197	65	submit	-
528	T049	J28	6	937	197	63	submit	-
528	T050	J28	6	937	197	65	submit	-
528	T051	J28	6	937	197	65	submit	-
528	T048	J28	6	937	197	65	start	M01
528	T049	J28	6	937	197	63	start	M01
528	T050	J28	6	937	197	65	start	M03
528	T051	J28	6	937	197	65	start	M03
545	T052	J29	4	312	6	18	submit	-
545	T052	J29	4	312	6	18	start	M02
547	T053	J30	4	1250	466	206	submit	-
547	T054	J30	4	1250	466	206	submit	-
547	T055	J30	4	1250	466	205	submit	-
547	T056	J30	4	1250	466	205	submit	-
547	T053	J30	4	1250	466	206	start	M03
547	T054	J30	4	1250	466	206	start	M07
547	T055	J30	4	1250	466	205	start	M07
547	T056	J30	4	1250	466	205	start	M07
563	T052	J29	4	312	6	18	finish	M02
591	T049	J28	6	937	197	63	finish	M01
593	T048	J28	6	937	197	65	finish	M01

time	task	job	priority	CPU	memory	run time	event	machine
593	T050	J28	6	937	197	65	finish	M03
593	T051	J28	6	937	197	65	finish	M03
601	T057	J31	4	1250	466	197	submit	-
601	T057	J31	4	1250	466	197	start	M01
602	T058	J32	4	1250	466	196	submit	-
602	T059	J32	4	1250	466	195	submit	-
602	T060	J32	4	1250	466	195	submit	-
602	T061	J32	4	1250	466	194	submit	-
602	T058	J32	4	1250	466	196	start	M03
602	T059	J32	4	1250	466	195	start	M03
602	T060	J32	4	1250	466	195	start	M07
602	T061	J32	4	1250	466	194	start	M07
610	T042	J25	6	625	280	132	finish	M01
626	T062	J33	8	2499	3339	860	submit	-
626	T063	J34	6	625	3180	27	submit	-
626	T063	J34	6	625	3180	27	start	M01
632	T046	J27	4	2499	3339	135	finish	M08
632	T062	J33	8	2499	3339	860	start	M08
634	T044	J27	4	2499	3339	137	finish	M01
640	T041	J24	8	6250	3107	179	finish	M05
641	T038	J21	9	6250	3180	230	finish	M02
646	T064	J35	0	6250	1244	178	submit	-
646	T064	J35	0	6250	1244	178	start	M02
647	T065	J36	0	6250	1244	179	submit	-
647	T065	J36	0	6250	1244	179	start	M05
648	T066	J37	9	6250	3180	200	submit	-
648	T064	J35	0	6250	1244	178	evict	M02
648	T066	J37	9	6250	3180	200	start	M02
649	T067	J38	9	4688	6360	536	submit	-
653	T063	J34	6	625	3180	27	finish	M01
653	T039	J22	1	1250	1590	353	evict	M01
653	T067	J38	9	4688	6360	536	start	M01
653	T039	J22	1	1250	1590	353	start	M08
659	T047	J27	4	2499	3339	138	finish	M03
668	T068	J39	8	6250	4773	797	submit	-
668	T069	J40	8	6250	1553	322	submit	-
685	T070	J41	4	6250	1553	304	submit	-
685	T071	J42	8	6250	1553	323	submit	-
687	T072	J43	4	6250	1553	319	submit	-
702	T035	J18	0	3125	311	354	finish	M02
703	T073	J44	8	6250	932	1099	submit	-
706	T074	J45	0	1250	16	24	submit	-
706	T074	J45	0	1250	16	24	start	M02
722	T075	J46	4	1250	398	44	submit	-
722	T075	J46	4	1250	398	44	start	M03
724	T076	J47	4	1250	466	168	submit	-
724	T077	J47	4	1250	466	167	submit	-
724	T078	J47	4	1250	466	164	submit	-
724	T079	J47	4	1250	466	168	submit	-
724	T076	J47	4	1250	466	168	start	M03
724	T077	J47	4	1250	466	167	start	M08
724	T078	J47	4	1250	466	164	start	M08
730	T074	J45	0	1250	16	24	finish	M02
745	T080	J48	8	2499	3339	897	submit	-
752	T055	J30	4	1250	466	205	finish	M07
752	T056	J30	4	1250	466	205	finish	M07
752	T079	J47	4	1250	466	168	start	M07

time	task	job	priority	CPU	memory	run time	event	machine
753	T053	J30	4	1250	466	206	finish	M03
753	T054	J30	4	1250	466	206	finish	M07
761	T045	J27	4	2499	3339	264	finish	M02
761	T068	J39	8	6250	4773	797	start	M02
763	T081	J49	4	1250	466	216	submit	-
763	T081	J49	4	1250	466	216	start	M03
766	T075	J46	4	1250	398	44	finish	M03
780	T082	J50	8	6250	4773	843	submit	-
796	T061	J32	4	1250	466	194	finish	M07
797	T059	J32	4	1250	466	195	finish	M03
797	T060	J32	4	1250	466	195	finish	M07
797	T080	J48	8	2499	3339	897	start	M03
798	T057	J31	4	1250	466	197	finish	M01
798	T058	J32	4	1250	466	196	finish	M03
805	T083	J51	8	1312	3107	1559	submit	-
822	T084	J52	4	1250	3107	1540	submit	-
822	T085	J52	4	1250	3107	1539	submit	-
822	T086	J52	4	1250	3107	1540	submit	-
822	T087	J52	4	1250	3107	99	submit	-
826	T065	J36	0	6250	1244	179	finish	M05
826	T069	J40	8	6250	1553	322	start	M05
848	T066	J37	9	6250	3180	200	finish	M02
848	T071	J42	8	6250	1553	323	start	M02
888	T078	J47	4	1250	466	164	finish	M08
891	T077	J47	4	1250	466	167	finish	M08
892	T076	J47	4	1250	466	168	finish	M03
895	T088	J53	4	3125	1125	45	submit	-
920	T079	J47	4	1250	466	168	finish	M07
920	T073	J44	8	6250	932	1099	start	M07
960	T089	J54	4	62	398	447	submit	-
960	T089	J54	4	62	398	447	start	M01
979	T081	J49	4	1250	466	216	finish	M03
979	T088	J53	4	3125	1125	45	start	M03
1006	T039	J22	1	1250	1590	353	finish	M08
1006	T090	J55	1	3125	795	6831	submit	-
1006	T091	J56	0	6250	466	45	submit	-
1006	T090	J55	1	3125	795	6831	start	M08
1009	T092	J57	0	3125	795	6827	submit	-
1012	T093	J58	0	625	387	91	submit	-
1012	T094	J58	0	625	387	92	submit	-
1012	T095	J58	0	625	387	93	submit	-
1012	T096	J58	0	625	387	102	submit	-
1012	T097	J58	0	625	387	90	submit	-
1012	T098	J58	0	625	387	92	submit	-
1012	T099	J58	0	625	387	98	submit	-
1012	T100	J58	0	625	387	99	submit	-
1012	T101	J58	0	625	387	89	submit	-
1012	T102	J58	0	625	387	188	submit	-
1012	T103	J58	0	625	387	118	submit	-
1012	T104	J58	0	625	387	112	submit	-
1012	T105	J58	0	625	387	112	submit	-
1012	T106	J58	0	625	387	93	submit	-
1012	T107	J58	0	625	387	90	submit	-
1012	T108	J58	0	625	387	108	submit	-
1012	T093	J58	0	625	387	91	start	M01
1012	T094	J58	0	625	387	92	start	M01
1012	T095	J58	0	625	387	93	start	M03

time	task	job	priority	CPU	memory	run time	event	machine
1012	T096	J58	0	625	387	102	start	M08
1014	T109	J59	8	6250	1553	347	submit	-
1014	T110	J60	4	625	155	40	submit	-
1024	T088	J53	4	3125	1125	45	finish	M03
1024	T110	J60	4	625	155	40	start	M03
1024	T097	J58	0	625	387	90	start	M03
1024	T098	J58	0	625	387	92	start	M03
1024	T099	J58	0	625	387	98	start	M03
1024	T100	J58	0	625	387	99	start	M03
1035	T111	J61	4	625	311	28	submit	-
1035	T112	J62	4	6250	1553	327	submit	-
1035	T113	J63	4	6250	1553	62	submit	-
1035	T114	J63	4	6250	1553	91	submit	-
1035	T115	J63	4	6250	1553	62	submit	-
1035	T116	J63	4	6250	1553	132	submit	-
1035	T117	J64	8	6250	1553	310	submit	-
1053	T118	J65	4	6250	1553	96	submit	-
1053	T119	J65	4	6250	1553	72	submit	-
1053	T120	J65	4	6250	1553	59	submit	-
1053	T121	J65	4	6250	1553	67	submit	-
1064	T110	J60	4	625	155	40	finish	M03
1064	T111	J61	4	625	311	28	start	M03
1092	T111	J61	4	625	311	28	finish	M03
1092	T101	J58	0	625	387	89	start	M03
1103	T093	J58	0	625	387	91	finish	M01
1103	T102	J58	0	625	387	188	start	M01
1104	T094	J58	0	625	387	92	finish	M01
1104	T103	J58	0	625	387	118	start	M01
1105	T095	J58	0	625	387	93	finish	M03
1105	T104	J58	0	625	387	112	start	M03
1114	T096	J58	0	625	387	102	finish	M08
1114	T097	J58	0	625	387	90	finish	M03
1114	T105	J58	0	625	387	112	start	M03
1114	T106	J58	0	625	387	93	start	M08
1116	T098	J58	0	625	387	92	finish	M03
1116	T107	J58	0	625	387	90	start	M03
1122	T099	J58	0	625	387	98	finish	M03
1122	T108	J58	0	625	387	108	start	M03
1123	T100	J58	0	625	387	99	finish	M03
1126	T122	J66	0	6250	1244	149	submit	-
1127	T123	J67	0	6250	1244	165	submit	-
1133	T124	J68	4	62	398	16	submit	-
1133	T124	J68	4	62	398	16	start	M01
1148	T069	J40	8	6250	1553	322	finish	M05
1148	T082	J50	8	6250	4773	843	start	M05
1149	T124	J68	4	62	398	16	finish	M01
1171	T071	J42	8	6250	1553	323	finish	M02
1171	T083	J51	8	1312	3107	1559	start	M02
1171	T084	J52	4	1250	3107	1540	start	M02
1171	T092	J57	0	3125	795	6827	start	M02
1181	T101	J58	0	625	387	89	finish	M03
1186	T125	J69	1	62	398	24	submit	-
1186	T125	J69	1	62	398	24	start	M01
1187	T126	J70	9	6250	1590	74	submit	-
1189	T067	J38	9	4688	6360	536	finish	M01
1189	T087	J52	4	1250	3107	99	start	M01

Cell window 01: Events

Time 359: machine M04 failed; no task was running on it.

Time 648: task T066 (priority 9) evicted task T064 (priority 0) from machine M02.

Time 653: task T067 (priority 9) evicted task T039 (priority 1) from machine M01.

Time 653: task T039 restarted on machine M08.

Longest waits (submission to first start, tasks started by the window end): task T082 waited 368 s (submitted at 780, first started at 1148); task T087 waited 367 s (submitted at 822, first started at 1189); task T083 waited 366 s (submitted at 805, first started at 1171); task T084 waited 349 s (submitted at 822, first started at 1171); task T073 waited 217 s (submitted at 703, first started at 920).

Cell window 01: Queue at window end

State after the events at time 1200.

Tasks running at time 1200 (remaining run time = seconds still to run after time 1200):

task	job	priority	CPU	memory	machine	started	remaining
T087	J52	4	1250	3107	M01	1189	88
T089	J54	4	62	398	M01	960	207
T102	J58	0	625	387	M01	1103	91
T103	J58	0	625	387	M01	1104	22
T125	J69	1	62	398	M01	1186	10
T068	J39	8	6250	4773	M02	761	358
T083	J51	8	1312	3107	M02	1171	1530
T084	J52	4	1250	3107	M02	1171	1511
T092	J57	0	3125	795	M02	1171	6798
T080	J48	8	2499	3339	M03	797	494
T104	J58	0	625	387	M03	1105	17
T105	J58	0	625	387	M03	1114	26
T107	J58	0	625	387	M03	1116	6
T108	J58	0	625	387	M03	1122	30
T082	J50	8	6250	4773	M05	1148	791
T043	J26	8	6250	4773	M06	478	810
T073	J44	8	6250	932	M07	920	819
T062	J33	8	2499	3339	M08	632	292
T090	J55	1	3125	795	M08	1006	6637
T106	J58	0	625	387	M08	1114	7

Tasks waiting at time 1200, in queue order (each will run its full run time once started):

task	job	priority	CPU	memory	submitted	run time
T126	J70	9	6250	1590	1187	74
T109	J59	8	6250	1553	1014	347
T117	J64	8	6250	1553	1035	310
T070	J41	4	6250	1553	685	304
T072	J43	4	6250	1553	687	319
T085	J52	4	1250	3107	822	1539
T086	J52	4	1250	3107	822	1540
T112	J62	4	6250	1553	1035	327
T113	J63	4	6250	1553	1035	62
T114	J63	4	6250	1553	1035	91
T115	J63	4	6250	1553	1035	62
T116	J63	4	6250	1553	1035	132
T118	J65	4	6250	1553	1053	96
T119	J65	4	6250	1553	1053	72
T120	J65	4	6250	1553	1053	59

task	job	priority	CPU	memory	submitted	run time
T121	J65	4	6250	1553	1053	67
T064	J35	0	6250	1244	646	178
T091	J56	0	6250	466	1006	45
T122	J66	0	6250	1244	1126	149
T123	J67	0	6250	1244	1127	165

Machine M04 is down at time 1200; it does not return.

Cell window 06

Times 0 to 1200. The cluster has 8 machines. 113 tasks of 32 jobs are submitted during the window.

Cell window 06: Machines

machine	CPU	memory	fails at	returns at
M01	6250	6244	-	-
M02	6250	3116	361	11295
M03	6250	6244	-	-
M04	6250	6244	-	-
M05	6250	6244	-	-
M06	6250	6244	-	-
M07	6250	3116	-	-
M08	6250	9363	-	-

Cell window 06: Task log

time	task	job	priority	CPU	memory	run time	event	machine
26	T001	J01	9	6250	1590	120	submit	-
26	T001	J01	9	6250	1590	120	start	M01
53	T002	J02	4	3125	1125	44	submit	-
53	T003	J03	4	1250	466	350	submit	-
53	T002	J02	4	3125	1125	44	start	M02
53	T003	J03	4	1250	466	350	start	M02
72	T004	J04	4	1250	466	274	submit	-
72	T005	J04	4	1250	466	312	submit	-
72	T006	J04	4	1250	466	317	submit	-
72	T007	J04	4	1250	466	296	submit	-
72	T004	J04	4	1250	466	274	start	M02
72	T005	J04	4	1250	466	312	start	M03
72	T006	J04	4	1250	466	317	start	M03
72	T007	J04	4	1250	466	296	start	M03
97	T002	J02	4	3125	1125	44	finish	M02
129	T008	J05	4	6250	3979	466	submit	-
129	T008	J05	4	6250	3979	466	start	M04
145	T009	J06	4	6250	3979	449	submit	-
145	T010	J06	4	6250	3979	448	submit	-
145	T011	J06	4	6250	3979	449	submit	-
145	T012	J06	4	6250	3979	448	submit	-
145	T009	J06	4	6250	3979	449	start	M05
145	T010	J06	4	6250	3979	448	start	M06
145	T011	J06	4	6250	3979	449	start	M08
146	T001	J01	9	6250	1590	120	finish	M01
146	T012	J06	4	6250	3979	448	start	M01
147	T013	J07	4	6250	1590	101	submit	-
147	T014	J07	4	6250	1590	102	submit	-
147	T015	J07	4	6250	1590	99	submit	-
147	T016	J07	4	6250	1590	111	submit	-
147	T013	J07	4	6250	1590	101	start	M07
178	T017	J08	4	3125	1125	47	submit	-
178	T017	J08	4	3125	1125	47	start	M02
210	T018	J09	9	4688	6360	783	submit	-
212	T019	J10	0	1250	197	252	submit	-
212	T019	J10	0	1250	197	252	start	M03
214	T020	J11	0	1250	204	156	submit	-
214	T021	J11	0	1250	204	206	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
214	T022	J11	0	1250	204	153	submit	-
214	T023	J11	0	1250	204	216	submit	-
214	T024	J11	0	1250	204	218	submit	-
214	T025	J11	0	1250	204	229	submit	-
214	T026	J11	0	1250	204	190	submit	-
214	T027	J11	0	1250	204	248	submit	-
214	T028	J11	0	1250	204	189	submit	-
214	T029	J11	0	1250	204	205	submit	-
214	T030	J11	0	1250	204	245	submit	-
214	T031	J11	0	1250	204	156	submit	-
214	T020	J11	0	1250	204	156	start	M03
225	T017	J08	4	3125	1125	47	finish	M02
225	T021	J11	0	1250	204	206	start	M02
225	T022	J11	0	1250	204	153	start	M02
225	T023	J11	0	1250	204	216	start	M02
248	T013	J07	4	6250	1590	101	finish	M07
248	T014	J07	4	6250	1590	102	start	M07
305	T032	J12	0	625	155	287	submit	-
305	T033	J12	0	625	155	250	submit	-
307	T034	J13	0	625	155	247	submit	-
307	T035	J13	0	625	155	248	submit	-
307	T036	J13	0	625	155	258	submit	-
307	T037	J13	0	625	155	259	submit	-
307	T038	J13	0	625	155	249	submit	-
307	T039	J13	0	625	155	251	submit	-
307	T040	J13	0	625	155	247	submit	-
307	T041	J13	0	625	155	257	submit	-
307	T042	J13	0	625	155	269	submit	-
307	T043	J13	0	625	155	245	submit	-
307	T044	J13	0	625	155	255	submit	-
307	T045	J13	0	625	155	279	submit	-
307	T046	J13	0	625	155	246	submit	-
307	T047	J13	0	625	155	246	submit	-
307	T048	J13	0	625	155	256	submit	-
307	T049	J13	0	625	155	260	submit	-
308	T050	J14	0	625	155	275	submit	-
308	T051	J14	0	625	155	278	submit	-
308	T052	J14	0	625	155	280	submit	-
308	T053	J14	0	625	155	281	submit	-
308	T054	J14	0	625	155	274	submit	-
308	T055	J14	0	625	155	279	submit	-
308	T056	J14	0	625	155	306	submit	-
308	T057	J14	0	625	155	277	submit	-
308	T058	J14	0	625	155	291	submit	-
308	T059	J14	0	625	155	282	submit	-
308	T060	J14	0	625	155	304	submit	-
308	T061	J14	0	625	155	300	submit	-
308	T062	J14	0	625	155	279	submit	-
308	T063	J14	0	625	155	283	submit	-
308	T064	J14	0	625	155	280	submit	-
308	T065	J14	0	625	155	275	submit	-
346	T004	J04	4	1250	466	274	finish	M02
346	T024	J11	0	1250	204	218	start	M02
350	T014	J07	4	6250	1590	102	finish	M07
350	T015	J07	4	6250	1590	99	start	M07
361	T003	J03	4	1250	466	350	fail	M02
361	T021	J11	0	1250	204	206	fail	M02

time	task	job	priority	CPU	memory	run time	event	machine
361	T022	J11	0	1250	204	153	fail	M02
361	T023	J11	0	1250	204	216	fail	M02
361	T024	J11	0	1250	204	218	fail	M02
368	T007	J04	4	1250	466	296	finish	M03
368	T003	J03	4	1250	466	350	start	M03
370	T020	J11	0	1250	204	156	finish	M03
370	T021	J11	0	1250	204	206	start	M03
374	T066	J15	8	625	16	35	submit	-
384	T005	J04	4	1250	466	312	finish	M03
384	T066	J15	8	625	16	35	start	M03
384	T032	J12	0	625	155	287	start	M03
389	T006	J04	4	1250	466	317	finish	M03
389	T022	J11	0	1250	204	153	start	M03
390	T067	J16	8	937	466	67	submit	-
393	T068	J17	4	625	197	66	submit	-
419	T066	J15	8	625	16	35	finish	M03
419	T068	J17	4	625	197	66	start	M03
449	T015	J07	4	6250	1590	99	finish	M07
449	T067	J16	8	937	466	67	start	M07
449	T023	J11	0	1250	204	216	start	M07
449	T024	J11	0	1250	204	218	start	M07
449	T025	J11	0	1250	204	229	start	M07
449	T026	J11	0	1250	204	190	start	M07
451	T069	J18	0	6250	1244	78	submit	-
453	T070	J19	1	1250	1590	375	submit	-
464	T019	J10	0	1250	197	252	finish	M03
464	T070	J19	1	1250	1590	375	start	M03
472	T071	J20	4	625	155	36	submit	-
473	T072	J21	4	625	155	30	submit	-
485	T068	J17	4	625	197	66	finish	M03
485	T071	J20	4	625	155	36	start	M03
515	T073	J22	1	1250	466	264	submit	-
516	T067	J16	8	937	466	67	finish	M07
516	T072	J21	4	625	155	30	start	M07
516	T033	J12	0	625	155	250	start	M07
518	T074	J23	0	625	219	261	submit	-
518	T075	J23	0	625	219	260	submit	-
518	T076	J23	0	625	219	258	submit	-
518	T077	J23	0	625	219	258	submit	-
518	T078	J23	0	625	219	259	submit	-
518	T079	J23	0	625	219	258	submit	-
518	T080	J23	0	625	219	257	submit	-
518	T081	J23	0	625	219	257	submit	-
518	T082	J23	0	625	219	257	submit	-
518	T083	J23	0	625	219	257	submit	-
518	T084	J23	0	625	219	260	submit	-
518	T085	J23	0	625	219	257	submit	-
518	T086	J23	0	625	219	260	submit	-
518	T087	J23	0	625	219	259	submit	-
518	T088	J23	0	625	219	259	submit	-
518	T089	J23	0	625	219	261	submit	-
521	T071	J20	4	625	155	36	finish	M03
521	T034	J13	0	625	155	247	start	M03
542	T022	J11	0	1250	204	153	finish	M03
542	T073	J22	1	1250	466	264	start	M03
546	T072	J21	4	625	155	30	finish	M07
546	T035	J13	0	625	155	248	start	M07

time	task	job	priority	CPU	memory	run time	event	machine
569	T090	J24	8	5090	466	824	submit	-
576	T021	J11	0	1250	204	206	finish	M03
576	T027	J11	0	1250	204	248	start	M03
593	T010	J06	4	6250	3979	448	finish	M06
593	T090	J24	8	5090	466	824	start	M06
593	T036	J13	0	625	155	258	start	M06
594	T009	J06	4	6250	3979	449	finish	M05
594	T011	J06	4	6250	3979	449	finish	M08
594	T012	J06	4	6250	3979	448	finish	M01
594	T018	J09	9	4688	6360	783	start	M08
594	T016	J07	4	6250	1590	111	start	M01
594	T028	J11	0	1250	204	189	start	M05
594	T029	J11	0	1250	204	205	start	M05
594	T030	J11	0	1250	204	245	start	M05
594	T031	J11	0	1250	204	156	start	M05
594	T037	J13	0	625	155	259	start	M05
594	T038	J13	0	625	155	249	start	M05
594	T039	J13	0	625	155	251	start	M08
594	T040	J13	0	625	155	247	start	M08
595	T008	J05	4	6250	3979	466	finish	M04
595	T041	J13	0	625	155	257	start	M04
595	T042	J13	0	625	155	269	start	M04
595	T043	J13	0	625	155	245	start	M04
595	T044	J13	0	625	155	255	start	M04
595	T045	J13	0	625	155	279	start	M04
595	T046	J13	0	625	155	246	start	M04
595	T047	J13	0	625	155	246	start	M04
595	T048	J13	0	625	155	256	start	M04
595	T049	J13	0	625	155	260	start	M04
595	T050	J14	0	625	155	275	start	M04
639	T026	J11	0	1250	204	190	finish	M07
639	T051	J14	0	625	155	278	start	M07
639	T052	J14	0	625	155	280	start	M07
665	T023	J11	0	1250	204	216	finish	M07
665	T053	J14	0	625	155	281	start	M07
665	T054	J14	0	625	155	274	start	M07
667	T024	J11	0	1250	204	218	finish	M07
667	T055	J14	0	625	155	279	start	M07
667	T056	J14	0	625	155	306	start	M07
671	T032	J12	0	625	155	287	finish	M03
671	T057	J14	0	625	155	277	start	M03
678	T025	J11	0	1250	204	229	finish	M07
678	T058	J14	0	625	155	291	start	M07
678	T059	J14	0	625	155	282	start	M07
705	T016	J07	4	6250	1590	111	finish	M01
705	T060	J14	0	625	155	304	start	M01
705	T061	J14	0	625	155	300	start	M01
705	T062	J14	0	625	155	279	start	M01
705	T063	J14	0	625	155	283	start	M01
705	T064	J14	0	625	155	280	start	M01
705	T065	J14	0	625	155	275	start	M01
705	T074	J23	0	625	219	261	start	M01
705	T075	J23	0	625	219	260	start	M01
705	T076	J23	0	625	219	258	start	M01
705	T077	J23	0	625	219	258	start	M01
718	T003	J03	4	1250	466	350	finish	M03
718	T078	J23	0	625	219	259	start	M03

time	task	job	priority	CPU	memory	run time	event	machine
718	T079	J23	0	625	219	258	start	M03
750	T031	J11	0	1250	204	156	finish	M05
750	T091	J25	0	1250	16	18	submit	-
750	T092	J26	9	6250	3180	330	submit	-
750	T077	J23	0	625	219	258	evict	M01
750	T076	J23	0	625	219	258	evict	M01
750	T075	J23	0	625	219	260	evict	M01
750	T074	J23	0	625	219	261	evict	M01
750	T065	J14	0	625	155	275	evict	M01
750	T064	J14	0	625	155	280	evict	M01
750	T063	J14	0	625	155	283	evict	M01
750	T062	J14	0	625	155	279	evict	M01
750	T061	J14	0	625	155	300	evict	M01
750	T060	J14	0	625	155	304	evict	M01
750	T092	J26	9	6250	3180	330	start	M01
750	T060	J14	0	625	155	304	start	M05
750	T061	J14	0	625	155	300	start	M05
766	T033	J12	0	625	155	250	finish	M07
766	T062	J14	0	625	155	279	start	M07
768	T034	J13	0	625	155	247	finish	M03
768	T063	J14	0	625	155	283	start	M03
783	T028	J11	0	1250	204	189	finish	M05
783	T064	J14	0	625	155	280	start	M05
783	T065	J14	0	625	155	275	start	M05
794	T035	J13	0	625	155	248	finish	M07
794	T074	J23	0	625	219	261	start	M07
799	T029	J11	0	1250	204	205	finish	M05
799	T075	J23	0	625	219	260	start	M05
799	T076	J23	0	625	219	258	start	M05
806	T073	J22	1	1250	466	264	finish	M03
806	T077	J23	0	625	219	258	start	M03
806	T080	J23	0	625	219	257	start	M03
824	T027	J11	0	1250	204	248	finish	M03
824	T081	J23	0	625	219	257	start	M03
824	T082	J23	0	625	219	257	start	M03
839	T030	J11	0	1250	204	245	finish	M05
839	T070	J19	1	1250	1590	375	finish	M03
839	T083	J23	0	625	219	257	start	M03
839	T084	J23	0	625	219	260	start	M03
839	T085	J23	0	625	219	257	start	M05
839	T086	J23	0	625	219	260	start	M05
840	T043	J13	0	625	155	245	finish	M04
841	T040	J13	0	625	155	247	finish	M08
841	T046	J13	0	625	155	246	finish	M04
841	T047	J13	0	625	155	246	finish	M04
841	T087	J23	0	625	219	259	start	M08
841	T091	J25	0	1250	16	18	start	M04
843	T038	J13	0	625	155	249	finish	M05
843	T088	J23	0	625	219	259	start	M05
845	T039	J13	0	625	155	251	finish	M08
845	T089	J23	0	625	219	261	start	M08
850	T044	J13	0	625	155	255	finish	M04
851	T036	J13	0	625	155	258	finish	M06
851	T048	J13	0	625	155	256	finish	M04
852	T041	J13	0	625	155	257	finish	M04
853	T037	J13	0	625	155	259	finish	M05
855	T049	J13	0	625	155	260	finish	M04

time	task	job	priority	CPU	memory	run time	event	machine
859	T091	J25	0	1250	16	18	finish	M04
864	T042	J13	0	625	155	269	finish	M04
870	T050	J14	0	625	155	275	finish	M04
874	T045	J13	0	625	155	279	finish	M04
874	T069	J18	0	6250	1244	78	start	M04
917	T051	J14	0	625	155	278	finish	M07
919	T052	J14	0	625	155	280	finish	M07
930	T093	J27	9	62	2	26	submit	-
930	T093	J27	9	62	2	26	start	M05
939	T054	J14	0	625	155	274	finish	M07
944	T094	J28	0	3125	311	2278	submit	-
944	T095	J28	0	3125	311	2498	submit	-
944	T096	J28	0	3125	311	2133	submit	-
944	T097	J28	0	3125	311	2452	submit	-
944	T098	J28	0	3125	311	2133	submit	-
944	T099	J28	0	3125	311	2098	submit	-
944	T100	J28	0	3125	311	2134	submit	-
944	T101	J28	0	3125	311	2227	submit	-
944	T102	J28	0	3125	311	2116	submit	-
944	T103	J28	0	3125	311	2427	submit	-
944	T104	J28	0	3125	311	2133	submit	-
944	T105	J28	0	3125	311	2094	submit	-
944	T106	J28	0	3125	311	2097	submit	-
944	T107	J28	0	3125	311	2102	submit	-
944	T108	J28	0	3125	311	2094	submit	-
944	T109	J28	0	3125	311	2115	submit	-
946	T053	J14	0	625	155	281	finish	M07
946	T055	J14	0	625	155	279	finish	M07
946	T094	J28	0	3125	311	2278	start	M07
948	T057	J14	0	625	155	277	finish	M03
952	T069	J18	0	6250	1244	78	finish	M04
952	T095	J28	0	3125	311	2498	start	M04
952	T096	J28	0	3125	311	2133	start	M04
956	T093	J27	9	62	2	26	finish	M05
960	T059	J14	0	625	155	282	finish	M07
969	T058	J14	0	625	155	291	finish	M07
973	T056	J14	0	625	155	306	finish	M07
976	T079	J23	0	625	219	258	finish	M03
977	T078	J23	0	625	219	259	finish	M03
990	T110	J29	4	3125	1224	142	submit	-
995	T111	J30	9	6250	1590	92	submit	-
995	T084	J23	0	625	219	260	evict	M03
995	T083	J23	0	625	219	257	evict	M03
995	T082	J23	0	625	219	257	evict	M03
995	T081	J23	0	625	219	257	evict	M03
995	T080	J23	0	625	219	257	evict	M03
995	T077	J23	0	625	219	258	evict	M03
995	T063	J14	0	625	155	283	evict	M03
995	T111	J30	9	6250	1590	92	start	M03
995	T063	J14	0	625	155	283	start	M05
995	T077	J23	0	625	219	258	start	M06
1006	T112	J31	8	625	466	34	submit	-
1006	T112	J31	8	625	466	34	start	M07
1040	T112	J31	8	625	466	34	finish	M07
1045	T062	J14	0	625	155	279	finish	M07
1050	T061	J14	0	625	155	300	finish	M05
1050	T080	J23	0	625	219	257	start	M05

time	task	job	priority	CPU	memory	run time	event	machine
1054	T060	J14	0	625	155	304	finish	M05
1054	T081	J23	0	625	219	257	start	M05
1055	T074	J23	0	625	219	261	finish	M07
1055	T110	J29	4	3125	1224	142	start	M07
1057	T076	J23	0	625	219	258	finish	M05
1057	T082	J23	0	625	219	257	start	M05
1058	T065	J14	0	625	155	275	finish	M05
1058	T083	J23	0	625	219	257	start	M05
1059	T075	J23	0	625	219	260	finish	M05
1059	T084	J23	0	625	219	260	start	M05
1063	T064	J14	0	625	155	280	finish	M05
1080	T092	J26	9	6250	3180	330	finish	M01
1080	T097	J28	0	3125	311	2452	start	M01
1080	T098	J28	0	3125	311	2133	start	M01
1087	T111	J30	9	6250	1590	92	finish	M03
1087	T099	J28	0	3125	311	2098	start	M03
1087	T100	J28	0	3125	311	2134	start	M03
1096	T085	J23	0	625	219	257	finish	M05
1099	T086	J23	0	625	219	260	finish	M05
1100	T087	J23	0	625	219	259	finish	M08
1102	T088	J23	0	625	219	259	finish	M05
1106	T089	J23	0	625	219	261	finish	M08
1172	T113	J32	9	4688	6360	579	submit	-
1197	T110	J29	4	3125	1224	142	finish	M07
1197	T101	J28	0	3125	311	2227	start	M07

Cell window 06: Events

Time 361: machine M02 failed; tasks T003, T021, T022, T023, T024 stopped and went back to the queue.

Time 368: task T003 restarted on machine M03.

Time 370: task T021 restarted on machine M03.

Time 389: task T022 restarted on machine M03.

Time 449: task T023 restarted on machine M07.

Time 449: task T024 restarted on machine M07.

Time 750: task T092 (priority 9) evicted task T060 (priority 0) from machine M01.

Time 750: task T092 (priority 9) evicted task T061 (priority 0) from machine M01.

Time 750: task T092 (priority 9) evicted task T062 (priority 0) from machine M01.

Time 750: task T092 (priority 9) evicted task T063 (priority 0) from machine M01.

Time 750: task T092 (priority 9) evicted task T064 (priority 0) from machine M01.

Time 750: task T092 (priority 9) evicted task T065 (priority 0) from machine M01.

Time 750: task T092 (priority 9) evicted task T074 (priority 0) from machine M01.

Time 750: task T092 (priority 9) evicted task T075 (priority 0) from machine M01.

Time 750: task T092 (priority 9) evicted task T076 (priority 0) from machine M01.

Time 750: task T092 (priority 9) evicted task T077 (priority 0) from machine M01.

Time 750: task T060 restarted on machine M05.

Time 750: task T061 restarted on machine M05.

Time 766: task T062 restarted on machine M07.

Time 768: task T063 restarted on machine M03.

Time 783: task T064 restarted on machine M05.

Time 783: task T065 restarted on machine M05.

Time 794: task T074 restarted on machine M07.

Time 799: task T075 restarted on machine M05.

Time 799: task T076 restarted on machine M05.

Time 806: task T077 restarted on machine M03.

Time 995: task T111 (priority 9) evicted task T063 (priority 0) from machine M03.

Time 995: task T111 (priority 9) evicted task T077 (priority 0) from machine M03.

Time 995: task T111 (priority 9) evicted task T080 (priority 0) from machine M03.

Time 995: task T111 (priority 9) evicted task T081 (priority 0) from machine M03.

Time 995: task T111 (priority 9) evicted task T082 (priority 0) from machine M03.

Time 995: task T111 (priority 9) evicted task T083 (priority 0) from machine M03.

Time 995: task T111 (priority 9) evicted task T084 (priority 0) from machine M03.

Time 995: task T063 restarted on machine M05.

Time 995: task T077 restarted on machine M06.

Time 1050: task T080 restarted on machine M05.

Time 1054: task T081 restarted on machine M05.

Time 1057: task T082 restarted on machine M05.

Time 1058: task T083 restarted on machine M05.

Time 1059: task T084 restarted on machine M05.

Longest waits (submission to first start, tasks started by the window end): task T016 waited 447 s (submitted at 147, first started at 594); task T069 waited 423 s (submitted at 451, first started at 874); task T060 waited 397 s (submitted at 308, first started at 705); task T061 waited 397 s (submitted at 308, first started at 705); task T062 waited 397 s (submitted at 308, first started at 705).

Cell window 06: Queue at window end

State after the events at time 1200.

Tasks running at time 1200 (remaining run time = seconds still to run after time 1200):

task	job	priority	CPU	memory	machine	started	remaining
T097	J28	0	3125	311	M01	1080	2332
T098	J28	0	3125	311	M01	1080	2013
T099	J28	0	3125	311	M03	1087	1985
T100	J28	0	3125	311	M03	1087	2021
T095	J28	0	3125	311	M04	952	2250
T096	J28	0	3125	311	M04	952	1885
T063	J14	0	625	155	M05	995	78
T080	J23	0	625	219	M05	1050	107
T081	J23	0	625	219	M05	1054	111
T082	J23	0	625	219	M05	1057	114
T083	J23	0	625	219	M05	1058	115
T084	J23	0	625	219	M05	1059	119
T077	J23	0	625	219	M06	995	53
T090	J24	8	5090	466	M06	593	217

task	job	priority	CPU	memory	machine	started	remaining
T094	J28	0	3125	311	M07	946	2024
T101	J28	0	3125	311	M07	1197	2224
T018	J09	9	4688	6360	M08	594	177

Tasks waiting at time 1200, in queue order (each will run its full run time once started):

task	job	priority	CPU	memory	submitted	run time
T113	J32	9	4688	6360	1172	579
T102	J28	0	3125	311	944	2116
T103	J28	0	3125	311	944	2427
T104	J28	0	3125	311	944	2133
T105	J28	0	3125	311	944	2094
T106	J28	0	3125	311	944	2097
T107	J28	0	3125	311	944	2102
T108	J28	0	3125	311	944	2094
T109	J28	0	3125	311	944	2115

Machine M02 is down at time 1200; it returns at time 11295.

Cell window 09

Times 0 to 1200. The cluster has 8 machines. 220 tasks of 96 jobs are submitted during the window.

Cell window 09: Machines

machine	CPU	memory	fails at	returns at
M01	6250	3116	492	never
M02	6250	6244	-	-
M03	6250	3116	-	-
M04	12500	12500	-	-
M05	6250	6244	-	-
M06	6250	6244	-	-
M07	6250	9363	-	-
M08	6250	6244	-	-

Cell window 09: Task log

time	task	job	priority	CPU	memory	run time	event	machine
2	T001	J01	0	625	219	229	submit	-
2	T002	J01	0	625	219	227	submit	-
2	T003	J01	0	625	219	213	submit	-
2	T004	J01	0	625	219	215	submit	-
2	T005	J01	0	625	219	217	submit	-
2	T006	J01	0	625	219	226	submit	-
2	T007	J01	0	625	219	226	submit	-
2	T008	J01	0	625	219	212	submit	-
2	T009	J01	0	625	219	216	submit	-
2	T010	J01	0	625	219	226	submit	-
2	T011	J01	0	625	219	215	submit	-
2	T012	J01	0	625	219	225	submit	-
2	T013	J01	0	625	219	216	submit	-
2	T014	J01	0	625	219	218	submit	-
2	T015	J01	0	625	219	227	submit	-
2	T016	J01	0	625	219	214	submit	-
2	T001	J01	0	625	219	229	start	M01
2	T002	J01	0	625	219	227	start	M01
2	T003	J01	0	625	219	213	start	M01
2	T004	J01	0	625	219	215	start	M01
2	T005	J01	0	625	219	217	start	M01
2	T006	J01	0	625	219	226	start	M01
2	T007	J01	0	625	219	226	start	M01
2	T008	J01	0	625	219	212	start	M01
2	T009	J01	0	625	219	216	start	M01
2	T010	J01	0	625	219	226	start	M01
2	T011	J01	0	625	219	215	start	M02
2	T012	J01	0	625	219	225	start	M02
2	T013	J01	0	625	219	216	start	M02
2	T014	J01	0	625	219	218	start	M02
2	T015	J01	0	625	219	227	start	M02
2	T016	J01	0	625	219	214	start	M02
31	T017	J02	4	1250	795	404	submit	-
31	T017	J02	4	1250	795	404	start	M02
54	T018	J03	9	6250	1590	104	submit	-
54	T018	J03	9	6250	1590	104	start	M03
94	T019	J04	6	937	284	154	submit	-
94	T020	J04	6	937	284	85	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
94	T021	J04	6	937	284	157	submit	-
94	T022	J04	6	937	284	96	submit	-
94	T019	J04	6	937	284	154	start	M02
94	T020	J04	6	937	284	85	start	M04
94	T021	J04	6	937	284	157	start	M04
94	T022	J04	6	937	284	96	start	M04
158	T018	J03	9	6250	1590	104	finish	M03
179	T020	J04	6	937	284	85	finish	M04
190	T022	J04	6	937	284	96	finish	M04
193	T023	J05	0	6250	1244	103	submit	-
193	T023	J05	0	6250	1244	103	start	M03
194	T024	J06	9	4688	6360	530	submit	-
194	T024	J06	9	4688	6360	530	start	M04
214	T008	J01	0	625	219	212	finish	M01
215	T003	J01	0	625	219	213	finish	M01
216	T016	J01	0	625	219	214	finish	M02
217	T004	J01	0	625	219	215	finish	M01
217	T011	J01	0	625	219	215	finish	M02
218	T009	J01	0	625	219	216	finish	M01
218	T013	J01	0	625	219	216	finish	M02
219	T005	J01	0	625	219	217	finish	M01
220	T014	J01	0	625	219	218	finish	M02
220	T025	J07	4	1250	466	315	submit	-
220	T025	J07	4	1250	466	315	start	M01
227	T012	J01	0	625	219	225	finish	M02
228	T006	J01	0	625	219	226	finish	M01
228	T007	J01	0	625	219	226	finish	M01
228	T010	J01	0	625	219	226	finish	M01
229	T002	J01	0	625	219	227	finish	M01
229	T015	J01	0	625	219	227	finish	M02
231	T001	J01	0	625	219	229	finish	M01
248	T019	J04	6	937	284	154	finish	M02
249	T026	J08	6	625	187	111	submit	-
249	T026	J08	6	625	187	111	start	M01
251	T021	J04	6	937	284	157	finish	M04
251	T027	J09	0	1250	16	37	submit	-
251	T027	J09	0	1250	16	37	start	M01
271	T028	J10	6	937	205	94	submit	-
271	T029	J10	6	937	205	93	submit	-
271	T030	J10	6	937	205	92	submit	-
271	T031	J10	6	937	205	91	submit	-
271	T028	J10	6	937	205	94	start	M01
271	T029	J10	6	937	205	93	start	M01
271	T030	J10	6	937	205	92	start	M01
271	T031	J10	6	937	205	91	start	M02
282	T032	J11	6	625	280	217	submit	-
282	T032	J11	6	625	280	217	start	M02
288	T027	J09	0	1250	16	37	finish	M01
296	T023	J05	0	6250	1244	103	finish	M03
297	T033	J12	1	1562	1553	237	submit	-
297	T033	J12	1	1562	1553	237	start	M01
311	T034	J13	1	1562	1553	149	submit	-
311	T034	J13	1	1562	1553	149	start	M02
315	T035	J14	6	937	204	135	submit	-
315	T036	J14	6	937	204	141	submit	-
315	T037	J14	6	937	204	128	submit	-
315	T038	J14	6	937	204	164	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
315	T039	J15	6	937	204	77	submit	-
315	T040	J15	6	937	204	76	submit	-
315	T041	J15	6	937	204	77	submit	-
315	T035	J14	6	937	204	135	start	M02
315	T036	J14	6	937	204	141	start	M02
315	T037	J14	6	937	204	128	start	M03
315	T038	J14	6	937	204	164	start	M03
315	T039	J15	6	937	204	77	start	M03
315	T040	J15	6	937	204	76	start	M03
315	T041	J15	6	937	204	77	start	M03
321	T042	J16	1	1562	1553	383	submit	-
321	T042	J16	1	1562	1553	383	start	M03
328	T043	J17	1	1562	1553	114	submit	-
328	T043	J17	1	1562	1553	114	start	M04
335	T044	J18	1	1562	1553	317	submit	-
335	T044	J18	1	1562	1553	317	start	M04
336	T045	J19	1	1250	1553	183	submit	-
336	T045	J19	1	1250	1553	183	start	M04
340	T046	J20	0	1250	216	182	submit	-
340	T046	J20	0	1250	216	182	start	M04
344	T047	J21	1	1250	1553	106	submit	-
344	T047	J21	1	1250	1553	106	start	M05
346	T048	J22	0	1112	211	104	submit	-
346	T048	J22	0	1112	211	104	start	M04
356	T049	J23	1	1250	1553	77	submit	-
356	T049	J23	1	1250	1553	77	start	M05
360	T026	J08	6	625	187	111	finish	M01
360	T050	J24	1	1250	1553	329	submit	-
360	T050	J24	1	1250	1553	329	start	M05
362	T031	J10	6	937	205	91	finish	M02
362	T051	J25	0	1250	197	327	submit	-
362	T051	J25	0	1250	197	327	start	M05
363	T030	J10	6	937	205	92	finish	M01
364	T029	J10	6	937	205	93	finish	M01
364	T052	J26	0	1250	208	184	submit	-
364	T053	J26	0	1250	208	181	submit	-
364	T054	J26	0	1250	208	203	submit	-
364	T055	J26	0	1250	208	224	submit	-
364	T056	J26	0	1250	208	239	submit	-
364	T057	J26	0	1250	208	218	submit	-
364	T058	J26	0	1250	208	323	submit	-
364	T059	J26	0	1250	208	163	submit	-
364	T060	J26	0	1250	208	324	submit	-
364	T061	J26	0	1250	208	319	submit	-
364	T062	J26	0	1250	208	161	submit	-
364	T063	J26	0	1250	208	179	submit	-
364	T064	J26	0	1250	208	163	submit	-
364	T065	J26	0	1250	208	160	submit	-
364	T066	J26	0	1250	208	161	submit	-
364	T067	J26	0	1250	208	321	submit	-
364	T052	J26	0	1250	208	184	start	M01
364	T053	J26	0	1250	208	181	start	M01
364	T054	J26	0	1250	208	203	start	M05
364	T055	J26	0	1250	208	224	start	M06
364	T056	J26	0	1250	208	239	start	M06
364	T057	J26	0	1250	208	218	start	M06
364	T058	J26	0	1250	208	323	start	M06

time	task	job	priority	CPU	memory	run time	event	machine
364	T059	J26	0	1250	208	163	start	M06
364	T060	J26	0	1250	208	324	start	M07
364	T061	J26	0	1250	208	319	start	M07
364	T062	J26	0	1250	208	161	start	M07
364	T063	J26	0	1250	208	179	start	M07
364	T064	J26	0	1250	208	163	start	M07
364	T065	J26	0	1250	208	160	start	M08
364	T066	J26	0	1250	208	161	start	M08
364	T067	J26	0	1250	208	321	start	M08
365	T028	J10	6	937	205	94	finish	M01
378	T068	J27	1	1250	1553	257	submit	-
378	T068	J27	1	1250	1553	257	start	M08
381	T069	J28	4	3125	1125	42	submit	-
391	T040	J15	6	937	204	76	finish	M03
392	T039	J15	6	937	204	77	finish	M03
392	T041	J15	6	937	204	77	finish	M03
394	T070	J29	6	625	280	303	submit	-
394	T070	J29	6	625	280	303	start	M01
433	T049	J23	1	1250	1553	77	finish	M05
435	T017	J02	4	1250	795	404	finish	M02
439	T071	J30	6	1250	676	204	submit	-
439	T072	J30	6	1250	676	200	submit	-
439	T073	J30	6	1250	676	221	submit	-
439	T074	J30	6	1250	676	230	submit	-
439	T075	J31	6	937	291	138	submit	-
439	T076	J31	6	937	291	120	submit	-
439	T077	J31	6	937	291	72	submit	-
439	T078	J31	6	937	291	154	submit	-
439	T071	J30	6	1250	676	204	start	M02
439	T072	J30	6	1250	676	200	start	M03
439	T073	J30	6	1250	676	221	start	M05
439	T074	J30	6	1250	676	230	start	M08
439	T075	J31	6	937	291	138	start	M02
439	T076	J31	6	937	291	120	start	M03
439	T077	J31	6	937	291	72	start	M04
442	T043	J17	1	1562	1553	114	finish	M04
442	T078	J31	6	937	291	154	start	M04
443	T037	J14	6	937	204	128	finish	M03
450	T035	J14	6	937	204	135	finish	M02
450	T047	J21	1	1250	1553	106	finish	M05
450	T048	J22	0	1112	211	104	finish	M04
456	T036	J14	6	937	204	141	finish	M02
460	T034	J13	1	1562	1553	149	finish	M02
460	T069	J28	4	3125	1125	42	start	M02
479	T038	J14	6	937	204	164	finish	M03
492	T025	J07	4	1250	466	315	fail	M01
492	T033	J12	1	1562	1553	237	fail	M01
492	T052	J26	0	1250	208	184	fail	M01
492	T053	J26	0	1250	208	181	fail	M01
492	T070	J29	6	625	280	303	fail	M01
492	T070	J29	6	625	280	303	start	M03
492	T025	J07	4	1250	466	315	start	M04
492	T052	J26	0	1250	208	184	start	M03
492	T053	J26	0	1250	208	181	start	M05
499	T032	J11	6	625	280	217	finish	M02
502	T069	J28	4	3125	1125	42	finish	M02
502	T033	J12	1	1562	1553	237	start	M02

time	task	job	priority	CPU	memory	run time	event	machine
510	T079	J32	4	1250	466	254	submit	-
510	T080	J32	4	1250	466	252	submit	-
510	T081	J32	4	1250	466	251	submit	-
510	T082	J32	4	1250	466	252	submit	-
510	T079	J32	4	1250	466	254	start	M02
510	T080	J32	4	1250	466	252	start	M02
511	T077	J31	6	937	291	72	finish	M04
511	T081	J32	4	1250	466	251	start	M04
519	T045	J19	1	1250	1553	183	finish	M04
519	T082	J32	4	1250	466	252	start	M04
522	T046	J20	0	1250	216	182	finish	M04
524	T065	J26	0	1250	208	160	finish	M08
525	T062	J26	0	1250	208	161	finish	M07
525	T066	J26	0	1250	208	161	finish	M08
527	T059	J26	0	1250	208	163	finish	M06
527	T064	J26	0	1250	208	163	finish	M07
543	T063	J26	0	1250	208	179	finish	M07
545	T083	J33	9	6250	3036	377	submit	-
545	T058	J26	0	1250	208	323	evict	M06
545	T057	J26	0	1250	208	218	evict	M06
545	T056	J26	0	1250	208	239	evict	M06
545	T055	J26	0	1250	208	224	evict	M06
545	T083	J33	9	6250	3036	377	start	M06
545	T055	J26	0	1250	208	224	start	M04
545	T056	J26	0	1250	208	239	start	M07
545	T057	J26	0	1250	208	218	start	M07
545	T058	J26	0	1250	208	323	start	M07
552	T084	J34	0	6250	466	45	submit	-
559	T076	J31	6	937	291	120	finish	M03
562	T085	J35	4	625	155	34	submit	-
562	T086	J36	4	625	311	21	submit	-
562	T085	J35	4	625	155	34	start	M03
562	T086	J36	4	625	311	21	start	M08
567	T054	J26	0	1250	208	203	finish	M05
577	T075	J31	6	937	291	138	finish	M02
583	T086	J36	4	625	311	21	finish	M08
596	T078	J31	6	937	291	154	finish	M04
596	T085	J35	4	625	155	34	finish	M03
625	T087	J37	8	4999	1590	4531	submit	-
627	T088	J38	4	4999	1590	4363	submit	-
627	T089	J38	4	4999	1590	4352	submit	-
627	T090	J38	4	4999	1590	4367	submit	-
627	T091	J38	4	4999	1590	4360	submit	-
635	T068	J27	1	1250	1553	257	finish	M08
639	T072	J30	6	1250	676	200	finish	M03
643	T071	J30	6	1250	676	204	finish	M02
652	T044	J18	1	1562	1553	317	finish	M04
660	T073	J30	6	1250	676	221	finish	M05
669	T074	J30	6	1250	676	230	finish	M08
669	T087	J37	8	4999	1590	4531	start	M08
673	T053	J26	0	1250	208	181	finish	M05
676	T052	J26	0	1250	208	184	finish	M03
683	T061	J26	0	1250	208	319	finish	M07
685	T067	J26	0	1250	208	321	finish	M08
688	T060	J26	0	1250	208	324	finish	M07
689	T050	J24	1	1250	1553	329	finish	M05
689	T051	J25	0	1250	197	327	finish	M05

time	task	job	priority	CPU	memory	run time	event	machine
689	T088	J38	4	4999	1590	4363	start	M05
704	T042	J16	1	1562	1553	383	finish	M03
704	T089	J38	4	4999	1590	4352	start	M03
724	T024	J06	9	4688	6360	530	finish	M04
724	T090	J38	4	4999	1590	4367	start	M04
726	T092	J39	9	6250	1590	74	submit	-
726	T058	J26	0	1250	208	323	evict	M07
726	T057	J26	0	1250	208	218	evict	M07
726	T056	J26	0	1250	208	239	evict	M07
726	T092	J39	9	6250	1590	74	start	M07
726	T056	J26	0	1250	208	239	start	M04
726	T057	J26	0	1250	208	218	start	M04
726	T058	J26	0	1250	208	323	start	M05
739	T033	J12	1	1562	1553	237	finish	M02
757	T093	J40	8	625	155	333	submit	-
757	T093	J40	8	625	155	333	start	M02
762	T080	J32	4	1250	466	252	finish	M02
762	T081	J32	4	1250	466	251	finish	M04
764	T079	J32	4	1250	466	254	finish	M02
764	T091	J38	4	4999	1590	4360	start	M02
769	T055	J26	0	1250	208	224	finish	M04
771	T082	J32	4	1250	466	252	finish	M04
781	T094	J41	1	1562	1553	165	submit	-
781	T094	J41	1	1562	1553	165	start	M04
788	T095	J42	1	1562	1553	468	submit	-
788	T095	J42	1	1562	1553	468	start	M04
794	T096	J43	1	1562	1553	246	submit	-
795	T070	J29	6	625	280	303	finish	M03
800	T092	J39	9	6250	1590	74	finish	M07
800	T096	J43	1	1562	1553	246	start	M07
803	T097	J44	1	1562	1553	391	submit	-
803	T097	J44	1	1562	1553	391	start	M07
807	T025	J07	4	1250	466	315	finish	M04
810	T098	J45	1	1562	1553	302	submit	-
810	T098	J45	1	1562	1553	302	start	M04
831	T099	J46	1	1250	1553	347	submit	-
831	T099	J46	1	1250	1553	347	start	M07
833	T100	J47	0	1250	258	343	submit	-
833	T100	J47	0	1250	258	343	start	M03
834	T101	J48	8	1312	466	59	submit	-
834	T102	J49	8	1312	466	138	submit	-
834	T101	J48	8	1312	466	59	start	M07
835	T103	J50	0	1250	197	226	submit	-
835	T104	J50	0	1250	197	131	submit	-
835	T105	J50	0	1250	197	340	submit	-
835	T103	J50	0	1250	197	226	start	M08
844	T106	J51	1	1250	1553	398	submit	-
846	T107	J52	0	1250	197	392	submit	-
847	T108	J53	1	1250	1553	251	submit	-
848	T109	J54	0	1250	197	111	submit	-
848	T110	J54	0	1250	197	389	submit	-
848	T111	J54	0	1250	197	151	submit	-
848	T112	J54	0	1250	197	177	submit	-
848	T113	J54	0	1250	197	122	submit	-
848	T114	J54	0	1250	197	142	submit	-
848	T115	J54	0	1250	197	285	submit	-
848	T116	J54	0	1250	197	385	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
848	T117	J54	0	1250	197	253	submit	-
848	T118	J54	0	1250	197	341	submit	-
848	T119	J54	0	1250	197	391	submit	-
848	T120	J54	0	1250	197	144	submit	-
848	T121	J54	0	1250	197	230	submit	-
848	T122	J54	0	1250	197	295	submit	-
848	T123	J54	0	1250	197	374	submit	-
848	T124	J54	0	1250	197	389	submit	-
849	T125	J55	0	1250	197	250	submit	-
851	T126	J56	0	1250	205	137	submit	-
851	T127	J56	0	1250	205	126	submit	-
851	T128	J56	0	1250	205	112	submit	-
851	T129	J56	0	1250	205	80	submit	-
851	T130	J56	0	1250	205	146	submit	-
851	T131	J56	0	1250	205	100	submit	-
851	T132	J56	0	1250	205	101	submit	-
851	T133	J56	0	1250	205	130	submit	-
851	T134	J56	0	1250	205	117	submit	-
851	T135	J56	0	1250	205	77	submit	-
851	T136	J56	0	1250	205	78	submit	-
851	T137	J56	0	1250	205	151	submit	-
851	T138	J56	0	1250	205	92	submit	-
851	T139	J56	0	1250	205	136	submit	-
851	T140	J56	0	1250	205	246	submit	-
851	T141	J56	0	1250	205	155	submit	-
851	T142	J57	9	6250	3180	214	submit	-
851	T057	J26	0	1250	208	218	evict	M04
851	T056	J26	0	1250	208	239	evict	M04
851	T098	J45	1	1562	1553	302	evict	M04
851	T095	J42	1	1562	1553	468	evict	M04
851	T094	J41	1	1562	1553	165	evict	M04
851	T142	J57	9	6250	3180	214	start	M04
851	T106	J51	1	1250	1553	398	start	M04
852	T143	J58	1	1250	1553	178	submit	-
854	T144	J59	6	1250	682	327	submit	-
854	T145	J59	6	1250	682	311	submit	-
854	T146	J59	6	1250	682	325	submit	-
854	T147	J59	6	1250	682	306	submit	-
854	T148	J60	0	1250	715	176	submit	-
856	T149	J61	1	1250	1553	78	submit	-
858	T150	J62	0	1250	197	76	submit	-
870	T151	J63	6	625	3180	29	submit	-
870	T151	J63	6	625	3180	29	start	M02
888	T152	J64	6	6250	622	136	submit	-
888	T153	J65	6	625	280	45	submit	-
893	T101	J48	8	1312	466	59	finish	M07
893	T102	J49	8	1312	466	138	start	M07
899	T151	J63	6	625	3180	29	finish	M02
899	T153	J65	6	625	280	45	start	M02
913	T154	J66	9	4688	6360	577	submit	-
913	T099	J46	1	1250	1553	347	evict	M07
913	T097	J44	1	1562	1553	391	evict	M07
913	T096	J43	1	1562	1553	246	evict	M07
913	T154	J66	9	4688	6360	577	start	M07
922	T083	J33	9	6250	3036	377	finish	M06
922	T144	J59	6	1250	682	327	start	M06
922	T145	J59	6	1250	682	311	start	M06

time	task	job	priority	CPU	memory	run time	event	machine
922	T146	J59	6	1250	682	325	start	M06
922	T147	J59	6	1250	682	306	start	M06
922	T099	J46	1	1250	1553	347	start	M06
944	T153	J65	6	625	280	45	finish	M02
994	T155	J67	6	625	187	30	submit	-
994	T156	J68	4	3125	1224	57	submit	-
994	T155	J67	6	625	187	30	start	M02
996	T157	J69	4	625	197	64	submit	-
1010	T158	J70	1	1562	1553	118	submit	-
1024	T155	J67	6	625	187	30	finish	M02
1024	T157	J69	4	625	197	64	start	M02
1031	T102	J49	8	1312	466	138	finish	M07
1031	T094	J41	1	1562	1553	165	start	M07
1047	T159	J71	4	62	398	55	submit	-
1047	T160	J72	8	1874	777	581	submit	-
1049	T058	J26	0	1250	208	323	finish	M05
1049	T159	J71	4	62	398	55	start	M05
1061	T103	J50	0	1250	197	226	finish	M08
1061	T108	J53	1	1250	1553	251	start	M08
1062	T161	J73	1	1250	1553	50	submit	-
1064	T162	J74	0	1250	197	48	submit	-
1065	T142	J57	9	6250	3180	214	finish	M04
1065	T160	J72	8	1874	777	581	start	M04
1065	T156	J68	4	3125	1224	57	start	M04
1065	T143	J58	1	1250	1553	178	start	M04
1088	T157	J69	4	625	197	64	finish	M02
1090	T093	J40	8	625	155	333	finish	M02
1090	T149	J61	1	1250	1553	78	start	M02
1093	T163	J75	4	625	217	67	submit	-
1093	T164	J75	4	625	217	67	submit	-
1093	T165	J75	4	625	217	72	submit	-
1093	T166	J75	4	625	217	67	submit	-
1093	T163	J75	4	625	217	67	start	M05
1098	T167	J76	1	1562	1553	365	submit	-
1104	T159	J71	4	62	398	55	finish	M05
1104	T168	J77	8	1250	777	70	submit	-
1104	T164	J75	4	625	217	67	start	M05
1122	T156	J68	4	3125	1224	57	finish	M04
1122	T168	J77	8	1250	777	70	start	M04
1122	T165	J75	4	625	217	72	start	M04
1122	T166	J75	4	625	217	67	start	M04
1131	T169	J78	1	1639	1553	313	submit	-
1133	T170	J79	0	736	317	311	submit	-
1135	T171	J80	4	1250	466	235	submit	-
1135	T172	J80	4	1250	466	242	submit	-
1135	T173	J80	4	1250	466	246	submit	-
1135	T174	J80	4	1250	466	250	submit	-
1135	T175	J81	0	1250	835	148	submit	-
1135	T176	J81	0	1250	835	200	submit	-
1135	T177	J81	0	1250	835	218	submit	-
1135	T178	J81	0	1250	835	167	submit	-
1135	T179	J81	0	1250	835	143	submit	-
1135	T180	J81	0	1250	835	175	submit	-
1135	T181	J81	0	1250	835	196	submit	-
1135	T182	J81	0	1250	835	136	submit	-
1135	T183	J81	0	1250	835	199	submit	-
1135	T184	J81	0	1250	835	125	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
1135	T185	J81	0	1250	835	147	submit	-
1135	T186	J81	0	1250	835	217	submit	-
1135	T187	J81	0	1250	835	308	submit	-
1135	T188	J81	0	1250	835	234	submit	-
1135	T189	J81	0	1250	835	144	submit	-
1135	T190	J81	0	1250	835	306	submit	-
1138	T191	J82	1	1562	1553	105	submit	-
1145	T192	J83	1	1562	1553	102	submit	-
1146	T193	J84	8	1250	1166	460	submit	-
1154	T194	J85	0	6250	466	44	submit	-
1158	T195	J86	1	1250	1590	432	submit	-
1160	T163	J75	4	625	217	67	finish	M05
1162	T196	J87	1	3125	2386	245	submit	-
1164	T197	J88	4	625	311	240	submit	-
1164	T198	J89	4	625	155	40	submit	-
1164	T197	J88	4	625	311	240	start	M04
1164	T198	J89	4	625	155	40	start	M05
1168	T149	J61	1	1250	1553	78	finish	M02
1168	T193	J84	8	1250	1166	460	start	M02
1171	T164	J75	4	625	217	67	finish	M05
1176	T100	J47	0	1250	258	343	finish	M03
1176	T171	J80	4	1250	466	235	start	M03
1179	T199	J90	1	1250	1553	53	submit	-
1181	T200	J91	0	1250	197	49	submit	-
1182	T201	J92	8	6250	1590	1278	submit	-
1186	T202	J93	1	1250	1553	46	submit	-
1188	T203	J94	0	1250	197	44	submit	-
1189	T166	J75	4	625	217	67	finish	M04
1192	T168	J77	8	1250	777	70	finish	M04
1192	T172	J80	4	1250	466	242	start	M04
1194	T165	J75	4	625	217	72	finish	M04
1194	T173	J80	4	1250	466	246	start	M04
1195	T204	J95	1	1250	466	188	submit	-
1196	T094	J41	1	1562	1553	165	finish	M07
1196	T174	J80	4	1250	466	250	start	M07
1198	T205	J96	0	625	219	186	submit	-
1198	T206	J96	0	625	219	170	submit	-
1198	T207	J96	0	625	219	170	submit	-
1198	T208	J96	0	625	219	186	submit	-
1198	T209	J96	0	625	219	174	submit	-
1198	T210	J96	0	625	219	170	submit	-
1198	T211	J96	0	625	219	176	submit	-
1198	T212	J96	0	625	219	175	submit	-
1198	T213	J96	0	625	219	186	submit	-
1198	T214	J96	0	625	219	182	submit	-
1198	T215	J96	0	625	219	182	submit	-
1198	T216	J96	0	625	219	183	submit	-
1198	T217	J96	0	625	219	183	submit	-
1198	T218	J96	0	625	219	170	submit	-
1198	T219	J96	0	625	219	170	submit	-
1198	T220	J96	0	625	219	174	submit	-
1198	T205	J96	0	625	219	186	start	M05

Cell window 09: Events

Time 492: machine M01 failed; tasks T025, T033, T052, T053, T070 stopped and went back to the queue.

Time 492: task T025 restarted on machine M04.

Time 492: task T052 restarted on machine M03.

Time 492: task T053 restarted on machine M05.

Time 492: task T070 restarted on machine M03.

Time 502: task T033 restarted on machine M02.

Time 545: task T083 (priority 9) evicted task T055 (priority 0) from machine M06.

Time 545: task T083 (priority 9) evicted task T056 (priority 0) from machine M06.

Time 545: task T083 (priority 9) evicted task T057 (priority 0) from machine M06.

Time 545: task T083 (priority 9) evicted task T058 (priority 0) from machine M06.

Time 545: task T055 restarted on machine M04.

Time 545: task T056 restarted on machine M07.

Time 545: task T057 restarted on machine M07.

Time 545: task T058 restarted on machine M07.

Time 726: task T092 (priority 9) evicted task T056 (priority 0) from machine M07.

Time 726: task T092 (priority 9) evicted task T057 (priority 0) from machine M07.

Time 726: task T092 (priority 9) evicted task T058 (priority 0) from machine M07.

Time 726: task T056 restarted on machine M04.

Time 726: task T057 restarted on machine M04.

Time 726: task T058 restarted on machine M05.

Time 851: task T142 (priority 9) evicted task T056 (priority 0) from machine M04.

Time 851: task T142 (priority 9) evicted task T057 (priority 0) from machine M04.

Time 851: task T142 (priority 9) evicted task T094 (priority 1) from machine M04.

Time 851: task T142 (priority 9) evicted task T095 (priority 1) from machine M04.

Time 851: task T142 (priority 9) evicted task T098 (priority 1) from machine M04.

Time 913: task T154 (priority 9) evicted task T096 (priority 1) from machine M07.

Time 913: task T154 (priority 9) evicted task T097 (priority 1) from machine M07.

Time 913: task T154 (priority 9) evicted task T099 (priority 1) from machine M07.

Time 922: task T099 restarted on machine M06.

Time 1031: task T094 restarted on machine M07.

Longest waits (submission to first start, tasks started by the window end): task T149 waited 234 s (submitted at 856, first started at 1090); task T108 waited 214 s (submitted at 847, first started at 1061); task T143 waited 213 s (submitted at 852, first started at 1065); task T091 waited 137 s (submitted at 627, first started at 764); task T090 waited 97 s (submitted at 627, first started at 724).

Cell window 09: Queue at window end

State after the events at time 1200.

Tasks running at time 1200 (remaining run time = seconds still to run after time 1200):

task	job	priority	CPU	memory	machine	started	remaining
T091	J38	4	4999	1590	M02	764	3924
T193	J84	8	1250	1166	M02	1168	428

task	job	priority	CPU	memory	machine	started	remaining
T089	J38	4	4999	1590	M03	704	3856
T171	J80	4	1250	466	M03	1176	211
T090	J38	4	4999	1590	M04	724	3891
T106	J51	1	1250	1553	M04	851	49
T143	J58	1	1250	1553	M04	1065	43
T160	J72	8	1874	777	M04	1065	446
T172	J80	4	1250	466	M04	1192	234
T173	J80	4	1250	466	M04	1194	240
T197	J88	4	625	311	M04	1164	204
T088	J38	4	4999	1590	M05	689	3852
T198	J89	4	625	155	M05	1164	4
T205	J96	0	625	219	M05	1198	184
T099	J46	1	1250	1553	M06	922	69
T144	J59	6	1250	682	M06	922	49
T145	J59	6	1250	682	M06	922	33
T146	J59	6	1250	682	M06	922	47
T147	J59	6	1250	682	M06	922	28
T154	J66	9	4688	6360	M07	913	290
T174	J80	4	1250	466	M07	1196	246
T087	J37	8	4999	1590	M08	669	4000
T108	J53	1	1250	1553	M08	1061	112

Tasks waiting at time 1200, in queue order (each will run its full run time once started):

task	job	priority	CPU	memory	submitted	run time
T201	J92	8	6250	1590	1182	1278
T152	J64	6	6250	622	888	136
T095	J42	1	1562	1553	788	468
T096	J43	1	1562	1553	794	246
T097	J44	1	1562	1553	803	391
T098	J45	1	1562	1553	810	302
T158	J70	1	1562	1553	1010	118
T161	J73	1	1250	1553	1062	50
T167	J76	1	1562	1553	1098	365
T169	J78	1	1639	1553	1131	313
T191	J82	1	1562	1553	1138	105
T192	J83	1	1562	1553	1145	102
T195	J86	1	1250	1590	1158	432
T196	J87	1	3125	2386	1162	245
T199	J90	1	1250	1553	1179	53
T202	J93	1	1250	1553	1186	46
T204	J95	1	1250	466	1195	188
T056	J26	0	1250	208	364	239
T057	J26	0	1250	208	364	218
T084	J34	0	6250	466	552	45
T104	J50	0	1250	197	835	131
T105	J50	0	1250	197	835	340
T107	J52	0	1250	197	846	392
T109	J54	0	1250	197	848	111
T110	J54	0	1250	197	848	389
T111	J54	0	1250	197	848	151
T112	J54	0	1250	197	848	177
T113	J54	0	1250	197	848	122
T114	J54	0	1250	197	848	142
T115	J54	0	1250	197	848	285
T116	J54	0	1250	197	848	385
T117	J54	0	1250	197	848	253
T118	J54	0	1250	197	848	341

task	job	priority	CPU	memory	submitted	run time
T119	J54	0	1250	197	848	391
T120	J54	0	1250	197	848	144
T121	J54	0	1250	197	848	230
T122	J54	0	1250	197	848	295
T123	J54	0	1250	197	848	374
T124	J54	0	1250	197	848	389
T125	J55	0	1250	197	849	250
T126	J56	0	1250	205	851	137
T127	J56	0	1250	205	851	126
T128	J56	0	1250	205	851	112
T129	J56	0	1250	205	851	80
T130	J56	0	1250	205	851	146
T131	J56	0	1250	205	851	100
T132	J56	0	1250	205	851	101
T133	J56	0	1250	205	851	130
T134	J56	0	1250	205	851	117
T135	J56	0	1250	205	851	77
T136	J56	0	1250	205	851	78
T137	J56	0	1250	205	851	151
T138	J56	0	1250	205	851	92
T139	J56	0	1250	205	851	136
T140	J56	0	1250	205	851	246
T141	J56	0	1250	205	851	155
T148	J60	0	1250	715	854	176
T150	J62	0	1250	197	858	76
T162	J74	0	1250	197	1064	48
T170	J79	0	736	317	1133	311
T175	J81	0	1250	835	1135	148
T176	J81	0	1250	835	1135	200
T177	J81	0	1250	835	1135	218
T178	J81	0	1250	835	1135	167
T179	J81	0	1250	835	1135	143
T180	J81	0	1250	835	1135	175
T181	J81	0	1250	835	1135	196
T182	J81	0	1250	835	1135	136
T183	J81	0	1250	835	1135	199
T184	J81	0	1250	835	1135	125
T185	J81	0	1250	835	1135	147
T186	J81	0	1250	835	1135	217
T187	J81	0	1250	835	1135	308
T188	J81	0	1250	835	1135	234
T189	J81	0	1250	835	1135	144
T190	J81	0	1250	835	1135	306
T194	J85	0	6250	466	1154	44
T200	J91	0	1250	197	1181	49
T203	J94	0	1250	197	1188	44
T206	J96	0	625	219	1198	170
T207	J96	0	625	219	1198	170
T208	J96	0	625	219	1198	186
T209	J96	0	625	219	1198	174
T210	J96	0	625	219	1198	170
T211	J96	0	625	219	1198	176
T212	J96	0	625	219	1198	175
T213	J96	0	625	219	1198	186
T214	J96	0	625	219	1198	182
T215	J96	0	625	219	1198	182
T216	J96	0	625	219	1198	183

task	job	priority	CPU	memory	submitted	run time
T217	J96	0	625	219	1198	183
T218	J96	0	625	219	1198	170
T219	J96	0	625	219	1198	170
T220	J96	0	625	219	1198	174

Machine M01 is down at time 1200; it does not return.

Cell window 28

Times 0 to 1200. The cluster has 8 machines. 114 tasks of 46 jobs are submitted during the window.

Cell window 28: Machines

machine	CPU	memory	fails at	returns at
M01	6250	6244	517	never
M02	6250	9363	-	-
M03	6250	3116	-	-
M04	6250	3116	-	-
M05	6250	3116	-	-
M06	6250	6244	-	-
M07	6250	3116	-	-
M08	6250	3116	-	-

Cell window 28: Task log

time	task	job	priority	CPU	memory	run time	event	machine
17	T001	J01	1	625	311	53	submit	-
17	T001	J01	1	625	311	53	start	M01
70	T001	J01	1	625	311	53	finish	M01
88	T002	J02	4	2234	3122	269	submit	-
88	T003	J02	4	2234	3122	269	submit	-
88	T004	J02	4	2234	3122	264	submit	-
88	T005	J02	4	2234	3122	260	submit	-
88	T002	J02	4	2234	3122	269	start	M01
88	T003	J02	4	2234	3122	269	start	M01
88	T004	J02	4	2234	3122	264	start	M02
88	T005	J02	4	2234	3122	260	start	M02
117	T006	J03	0	3125	466	273	submit	-
117	T006	J03	0	3125	466	273	start	M03
149	T007	J04	0	2594	197	617	submit	-
149	T008	J04	0	2594	197	617	submit	-
149	T009	J04	0	2594	197	616	submit	-
149	T010	J04	0	2594	197	610	submit	-
149	T011	J04	0	2594	197	616	submit	-
149	T012	J04	0	2594	197	608	submit	-
149	T013	J04	0	2594	197	617	submit	-
149	T014	J04	0	2594	197	609	submit	-
149	T015	J04	0	2594	197	609	submit	-
149	T016	J04	0	2594	197	616	submit	-
149	T007	J04	0	2594	197	617	start	M03
149	T008	J04	0	2594	197	617	start	M04
149	T009	J04	0	2594	197	616	start	M04
149	T010	J04	0	2594	197	610	start	M05
149	T011	J04	0	2594	197	616	start	M05
149	T012	J04	0	2594	197	608	start	M06
149	T013	J04	0	2594	197	617	start	M06
149	T014	J04	0	2594	197	609	start	M07
149	T015	J04	0	2594	197	609	start	M07
149	T016	J04	0	2594	197	616	start	M08
166	T017	J05	8	3125	795	244	submit	-
166	T018	J06	8	625	155	315	submit	-
166	T017	J05	8	3125	795	244	start	M08
166	T018	J06	8	625	155	315	start	M02
194	T019	J07	4	3125	777	47	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
299	T020	J08	1	3748	466	203	submit	-
312	T021	J09	4	3125	1224	37	submit	-
312	T022	J10	4	3125	224	25	submit	-
348	T005	J02	4	2234	3122	260	finish	M02
348	T019	J07	4	3125	777	47	start	M02
352	T004	J02	4	2234	3122	264	finish	M02
357	T002	J02	4	2234	3122	269	finish	M01
357	T003	J02	4	2234	3122	269	finish	M01
357	T021	J09	4	3125	1224	37	start	M01
357	T022	J10	4	3125	224	25	start	M01
367	T023	J11	4	62	398	18	submit	-
367	T023	J11	4	62	398	18	start	M02
382	T022	J10	4	3125	224	25	finish	M01
385	T023	J11	4	62	398	18	finish	M02
390	T006	J03	0	3125	466	273	finish	M03
394	T021	J09	4	3125	1224	37	finish	M01
394	T020	J08	1	3748	466	203	start	M01
395	T019	J07	4	3125	777	47	finish	M02
410	T017	J05	8	3125	795	244	finish	M08
440	T024	J12	8	625	155	120	submit	-
440	T024	J12	8	625	155	120	start	M01
455	T025	J13	8	1250	777	59	submit	-
455	T025	J13	8	1250	777	59	start	M01
477	T026	J14	0	6250	466	34	submit	-
479	T027	J15	9	6250	233	103	submit	-
479	T007	J04	0	2594	197	617	evict	M03
479	T027	J15	9	6250	233	103	start	M03
479	T007	J04	0	2594	197	617	start	M02
481	T018	J06	8	625	155	315	finish	M02
481	T028	J16	4	3125	1125	31	submit	-
481	T029	J17	4	625	155	32	submit	-
481	T028	J16	4	3125	1125	31	start	M02
481	T029	J17	4	625	155	32	start	M01
482	T030	J18	9	6250	1590	75	submit	-
482	T009	J04	0	2594	197	616	evict	M04
482	T008	J04	0	2594	197	617	evict	M04
482	T030	J18	9	6250	1590	75	start	M04
482	T008	J04	0	2594	197	617	start	M08
496	T031	J19	4	3125	448	26	submit	-
496	T032	J20	4	625	311	194	submit	-
496	T032	J20	4	625	311	194	start	M05
512	T028	J16	4	3125	1125	31	finish	M02
512	T031	J19	4	3125	448	26	start	M02
513	T029	J17	4	625	155	32	finish	M01
514	T025	J13	8	1250	777	59	finish	M01
517	T020	J08	1	3748	466	203	fail	M01
517	T024	J12	8	625	155	120	fail	M01
517	T024	J12	8	625	155	120	start	M06
526	T033	J21	0	625	242	149	submit	-
526	T034	J21	0	625	242	149	submit	-
526	T035	J21	0	625	242	148	submit	-
526	T036	J21	0	625	242	147	submit	-
526	T037	J21	0	625	242	149	submit	-
526	T038	J21	0	625	242	147	submit	-
526	T039	J21	0	625	242	148	submit	-
526	T040	J21	0	625	242	148	submit	-
526	T041	J21	0	625	242	148	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
526	T042	J21	0	625	242	148	submit	-
526	T043	J21	0	625	242	149	submit	-
526	T044	J21	0	625	242	147	submit	-
526	T045	J21	0	625	242	148	submit	-
526	T046	J21	0	625	242	148	submit	-
526	T047	J21	0	625	242	149	submit	-
526	T048	J21	0	625	242	148	submit	-
526	T033	J21	0	625	242	149	start	M07
526	T034	J21	0	625	242	149	start	M08
538	T031	J19	4	3125	448	26	finish	M02
538	T009	J04	0	2594	197	616	start	M02
538	T035	J21	0	625	242	148	start	M02
540	T049	J22	1	3125	622	90	submit	-
548	T050	J23	4	3125	1125	33	submit	-
557	T030	J18	9	6250	1590	75	finish	M04
557	T050	J23	4	3125	1125	33	start	M04
557	T049	J22	1	3125	622	90	start	M04
582	T027	J15	9	6250	233	103	finish	M03
582	T020	J08	1	3748	466	203	start	M03
582	T036	J21	0	625	242	147	start	M03
582	T037	J21	0	625	242	149	start	M03
582	T038	J21	0	625	242	147	start	M03
582	T039	J21	0	625	242	148	start	M03
590	T050	J23	4	3125	1125	33	finish	M04
597	T051	J24	9	4688	6360	482	submit	-
597	T035	J21	0	625	242	148	evict	M02
597	T009	J04	0	2594	197	616	evict	M02
597	T007	J04	0	2594	197	617	evict	M02
597	T051	J24	9	4688	6360	482	start	M02
597	T007	J04	0	2594	197	617	start	M04
597	T035	J21	0	625	242	148	start	M02
597	T040	J21	0	625	242	148	start	M02
637	T024	J12	8	625	155	120	finish	M06
637	T041	J21	0	625	242	148	start	M06
647	T049	J22	1	3125	622	90	finish	M04
647	T009	J04	0	2594	197	616	start	M04
647	T042	J21	0	625	242	148	start	M04
675	T033	J21	0	625	242	149	finish	M07
675	T034	J21	0	625	242	149	finish	M08
675	T043	J21	0	625	242	149	start	M08
690	T032	J20	4	625	311	194	finish	M05
690	T044	J21	0	625	242	147	start	M05
693	T052	J25	1	1353	777	2052	submit	-
693	T045	J21	0	625	242	148	start	M07
695	T053	J26	0	1250	777	1211	submit	-
695	T054	J26	0	1250	777	2050	submit	-
695	T055	J26	0	1250	777	1113	submit	-
697	T056	J27	0	937	274	100	submit	-
697	T057	J27	0	937	274	159	submit	-
697	T058	J27	0	937	274	93	submit	-
697	T059	J27	0	937	274	92	submit	-
697	T060	J27	0	937	274	133	submit	-
697	T061	J27	0	937	274	87	submit	-
697	T062	J27	0	937	274	100	submit	-
697	T063	J27	0	937	274	110	submit	-
697	T064	J27	0	937	274	111	submit	-
697	T065	J27	0	937	274	93	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
697	T066	J27	0	937	274	144	submit	-
697	T067	J27	0	937	274	102	submit	-
697	T068	J27	0	937	274	121	submit	-
697	T069	J27	0	937	274	128	submit	-
697	T070	J27	0	937	274	100	submit	-
697	T071	J27	0	937	274	100	submit	-
718	T072	J28	0	6250	1244	36	submit	-
729	T036	J21	0	625	242	147	finish	M03
729	T038	J21	0	625	242	147	finish	M03
730	T039	J21	0	625	242	148	finish	M03
730	T052	J25	1	1353	777	2052	start	M03
731	T037	J21	0	625	242	149	finish	M03
731	T046	J21	0	625	242	148	start	M03
732	T073	J29	4	625	155	24	submit	-
745	T035	J21	0	625	242	148	finish	M02
745	T040	J21	0	625	242	148	finish	M02
745	T073	J29	4	625	155	24	start	M02
745	T047	J21	0	625	242	149	start	M02
756	T074	J30	1	1250	777	1135	submit	-
757	T012	J04	0	2594	197	608	finish	M06
757	T074	J30	1	1250	777	1135	start	M06
757	T048	J21	0	625	242	148	start	M06
757	T056	J27	0	937	274	100	start	M06
758	T014	J04	0	2594	197	609	finish	M07
758	T015	J04	0	2594	197	609	finish	M07
758	T053	J26	0	1250	777	1211	start	M07
758	T054	J26	0	1250	777	2050	start	M07
758	T055	J26	0	1250	777	1113	start	M07
758	T057	J27	0	937	274	159	start	M07
759	T010	J04	0	2594	197	610	finish	M05
759	T075	J31	8	937	1553	64	submit	-
759	T075	J31	8	937	1553	64	start	M05
765	T011	J04	0	2594	197	616	finish	M05
765	T016	J04	0	2594	197	616	finish	M08
765	T058	J27	0	937	274	93	start	M08
765	T059	J27	0	937	274	92	start	M08
765	T060	J27	0	937	274	133	start	M08
766	T013	J04	0	2594	197	617	finish	M06
766	T061	J27	0	937	274	87	start	M06
766	T062	J27	0	937	274	100	start	M06
766	T063	J27	0	937	274	110	start	M06
769	T073	J29	4	625	155	24	finish	M02
769	T064	J27	0	937	274	111	start	M02
773	T076	J32	4	4999	1590	2271	submit	-
773	T077	J33	4	4999	1590	2245	submit	-
774	T078	J34	4	656	197	247	submit	-
774	T079	J34	4	656	197	241	submit	-
774	T080	J34	4	656	197	240	submit	-
774	T081	J34	4	656	197	237	submit	-
774	T078	J34	4	656	197	247	start	M07
785	T020	J08	1	3748	466	203	finish	M03
785	T041	J21	0	625	242	148	finish	M06
785	T079	J34	4	656	197	241	start	M03
785	T080	J34	4	656	197	240	start	M03
785	T081	J34	4	656	197	237	start	M03
785	T065	J27	0	937	274	93	start	M03
785	T066	J27	0	937	274	144	start	M03

time	task	job	priority	CPU	memory	run time	event	machine
795	T042	J21	0	625	242	148	finish	M04
795	T067	J27	0	937	274	102	start	M04
796	T082	J35	1	6250	466	662	submit	-
799	T083	J36	8	6250	466	438	submit	-
814	T084	J37	4	6250	622	420	submit	-
814	T085	J37	4	6250	622	266	submit	-
814	T086	J37	4	6250	622	309	submit	-
814	T087	J37	4	6250	622	257	submit	-
823	T075	J31	8	937	1553	64	finish	M05
824	T043	J21	0	625	242	149	finish	M08
837	T044	J21	0	625	242	147	finish	M05
837	T083	J36	8	6250	466	438	start	M05
841	T045	J21	0	625	242	148	finish	M07
853	T061	J27	0	937	274	87	finish	M06
857	T056	J27	0	937	274	100	finish	M06
857	T059	J27	0	937	274	92	finish	M08
857	T068	J27	0	937	274	121	start	M08
858	T058	J27	0	937	274	93	finish	M08
858	T069	J27	0	937	274	128	start	M08
866	T062	J27	0	937	274	100	finish	M06
876	T063	J27	0	937	274	110	finish	M06
878	T065	J27	0	937	274	93	finish	M03
878	T070	J27	0	937	274	100	start	M03
879	T046	J21	0	625	242	148	finish	M03
879	T071	J27	0	937	274	100	start	M03
880	T064	J27	0	937	274	111	finish	M02
894	T047	J21	0	625	242	149	finish	M02
897	T067	J27	0	937	274	102	finish	M04
898	T060	J27	0	937	274	133	finish	M08
905	T048	J21	0	625	242	148	finish	M06
905	T076	J32	4	4999	1590	2271	start	M06
912	T088	J38	0	937	272	89	submit	-
912	T089	J38	0	937	272	83	submit	-
912	T090	J38	0	937	272	93	submit	-
912	T091	J38	0	937	272	79	submit	-
912	T092	J38	0	937	272	181	submit	-
912	T093	J38	0	937	272	102	submit	-
912	T094	J38	0	937	272	79	submit	-
912	T095	J38	0	937	272	83	submit	-
912	T096	J38	0	937	272	80	submit	-
912	T097	J38	0	937	272	98	submit	-
912	T098	J38	0	937	272	92	submit	-
912	T099	J38	0	937	272	83	submit	-
912	T100	J38	0	937	272	80	submit	-
912	T101	J38	0	937	272	80	submit	-
912	T102	J38	0	937	272	82	submit	-
912	T103	J38	0	937	272	92	submit	-
912	T088	J38	0	937	272	89	start	M02
912	T089	J38	0	937	272	83	start	M04
912	T090	J38	0	937	272	93	start	M08
917	T057	J27	0	937	274	159	finish	M07
917	T091	J38	0	937	272	79	start	M07
929	T066	J27	0	937	274	144	finish	M03
929	T092	J38	0	937	272	181	start	M03
954	T104	J39	8	6250	932	1356	submit	-
978	T068	J27	0	937	274	121	finish	M08
978	T070	J27	0	937	274	100	finish	M03

time	task	job	priority	CPU	memory	run time	event	machine
978	T093	J38	0	937	272	102	start	M03
978	T094	J38	0	937	272	79	start	M08
979	T071	J27	0	937	274	100	finish	M03
979	T095	J38	0	937	272	83	start	M03
986	T069	J27	0	937	274	128	finish	M08
986	T096	J38	0	937	272	80	start	M08
995	T089	J38	0	937	272	83	finish	M04
995	T097	J38	0	937	272	98	start	M04
996	T091	J38	0	937	272	79	finish	M07
996	T098	J38	0	937	272	92	start	M07
1001	T088	J38	0	937	272	89	finish	M02
1005	T090	J38	0	937	272	93	finish	M08
1005	T099	J38	0	937	272	83	start	M08
1013	T105	J40	4	625	208	99	submit	-
1013	T106	J40	4	625	208	98	submit	-
1013	T107	J40	4	625	208	99	submit	-
1013	T108	J40	4	625	208	99	submit	-
1013	T105	J40	4	625	208	99	start	M07
1013	T106	J40	4	625	208	98	start	M08
1016	T109	J41	0	3125	466	259	submit	-
1021	T078	J34	4	656	197	247	finish	M07
1021	T107	J40	4	625	208	99	start	M07
1022	T081	J34	4	656	197	237	finish	M03
1022	T108	J40	4	625	208	99	start	M03
1025	T080	J34	4	656	197	240	finish	M03
1026	T079	J34	4	656	197	241	finish	M03
1026	T100	J38	0	937	272	80	start	M03
1057	T094	J38	0	937	272	79	finish	M08
1057	T101	J38	0	937	272	80	start	M08
1062	T095	J38	0	937	272	83	finish	M03
1062	T102	J38	0	937	272	82	start	M03
1066	T096	J38	0	937	272	80	finish	M08
1066	T103	J38	0	937	272	92	start	M08
1079	T051	J24	9	4688	6360	482	finish	M02
1079	T104	J39	8	6250	932	1356	start	M02
1080	T093	J38	0	937	272	102	finish	M03
1081	T110	J42	9	6250	1590	69	submit	-
1081	T111	J43	4	625	155	24	submit	-
1081	T097	J38	0	937	272	98	evict	M04
1081	T009	J04	0	2594	197	616	evict	M04
1081	T007	J04	0	2594	197	617	evict	M04
1081	T110	J42	9	6250	1590	69	start	M04
1081	T111	J43	4	625	155	24	start	M03
1088	T098	J38	0	937	272	92	finish	M07
1088	T099	J38	0	937	272	83	finish	M08
1088	T097	J38	0	937	272	98	start	M07
1099	T008	J04	0	2594	197	617	finish	M08
1105	T111	J43	4	625	155	24	finish	M03
1106	T100	J38	0	937	272	80	finish	M03
1110	T092	J38	0	937	272	181	finish	M03
1110	T007	J04	0	2594	197	617	start	M03
1111	T106	J40	4	625	208	98	finish	M08
1112	T105	J40	4	625	208	99	finish	M07
1120	T107	J40	4	625	208	99	finish	M07
1121	T108	J40	4	625	208	99	finish	M03
1135	T112	J44	8	1312	466	74	submit	-
1135	T113	J45	8	937	466	67	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
1135	T112	J44	8	1312	466	74	start	M03
1135	T113	J45	8	937	466	67	start	M07
1137	T101	J38	0	937	272	80	finish	M08
1137	T077	J33	4	4999	1590	2245	start	M08
1144	T102	J38	0	937	272	82	finish	M03
1150	T110	J42	9	6250	1590	69	finish	M04
1150	T084	J37	4	6250	622	420	start	M04
1158	T103	J38	0	937	272	92	finish	M08
1186	T097	J38	0	937	272	98	finish	M07
1198	T114	J46	9	4688	6360	496	submit	-

Cell window 28: Events

Time 479: task T027 (priority 9) evicted task T007 (priority 0) from machine M03.

Time 479: task T007 restarted on machine M02.

Time 482: task T030 (priority 9) evicted task T008 (priority 0) from machine M04.

Time 482: task T030 (priority 9) evicted task T009 (priority 0) from machine M04.

Time 482: task T008 restarted on machine M08.

Time 517: machine M01 failed; tasks T020, T024 stopped and went back to the queue.

Time 517: task T024 restarted on machine M06.

Time 538: task T009 restarted on machine M02.

Time 582: task T020 restarted on machine M03.

Time 597: task T051 (priority 9) evicted task T007 (priority 0) from machine M02.

Time 597: task T051 (priority 9) evicted task T009 (priority 0) from machine M02.

Time 597: task T051 (priority 9) evicted task T035 (priority 0) from machine M02.

Time 597: task T007 restarted on machine M04.

Time 597: task T035 restarted on machine M02.

Time 647: task T009 restarted on machine M04.

Time 1081: task T110 (priority 9) evicted task T007 (priority 0) from machine M04.

Time 1081: task T110 (priority 9) evicted task T009 (priority 0) from machine M04.

Time 1081: task T110 (priority 9) evicted task T097 (priority 0) from machine M04.

Time 1088: task T097 restarted on machine M07.

Time 1110: task T007 restarted on machine M03.

Longest waits (submission to first start, tasks started by the window end): task T077 waited 364 s (submitted at 773, first started at 1137); task T084 waited 336 s (submitted at 814, first started at 1150); task T048 waited 231 s (submitted at 526, first started at 757); task T047 waited 219 s (submitted at 526, first started at 745); task T046 waited 205 s (submitted at 526, first started at 731).

Cell window 28: Queue at window end

State after the events at time 1200.

Tasks running at time 1200 (remaining run time = seconds still to run after time 1200):

task	job	priority	CPU	memory	machine	started	remaining
T104	J39	8	6250	932	M02	1079	1235
T007	J04	0	2594	197	M03	1110	527

task	job	priority	CPU	memory	machine	started	remaining
T052	J25	1	1353	777	M03	730	1582
T112	J44	8	1312	466	M03	1135	9
T084	J37	4	6250	622	M04	1150	370
T083	J36	8	6250	466	M05	837	75
T074	J30	1	1250	777	M06	757	692
T076	J32	4	4999	1590	M06	905	1976
T053	J26	0	1250	777	M07	758	769
T054	J26	0	1250	777	M07	758	1608
T055	J26	0	1250	777	M07	758	671
T113	J45	8	937	466	M07	1135	2
T077	J33	4	4999	1590	M08	1137	2182

Tasks waiting at time 1200, in queue order (each will run its full run time once started):

task	job	priority	CPU	memory	submitted	run time
T114	J46	9	4688	6360	1198	496
T085	J37	4	6250	622	814	266
T086	J37	4	6250	622	814	309
T087	J37	4	6250	622	814	257
T082	J35	1	6250	466	796	662
T009	J04	0	2594	197	149	616
T026	J14	0	6250	466	477	34
T072	J28	0	6250	1244	718	36
T109	J41	0	3125	466	1016	259

Machine M01 is down at time 1200; it does not return.

Cell window 29

Times 0 to 1200. The cluster has 8 machines. 146 tasks of 63 jobs are submitted during the window.

Cell window 29: Machines

machine	CPU	memory	fails at	returns at
M01	6250	3116	-	-
M02	6250	6244	-	-
M03	6250	3116	-	-
M04	6250	6244	-	-
M05	6250	6244	-	-
M06	6250	9363	394	1347
M07	6250	3116	-	-
M08	6250	3116	-	-

Cell window 29: Task log

time	task	job	priority	CPU	memory	run time	event	machine
4	T001	J01	0	625	242	149	submit	-
4	T002	J01	0	625	242	149	submit	-
4	T003	J01	0	625	242	148	submit	-
4	T004	J01	0	625	242	147	submit	-
4	T005	J01	0	625	242	149	submit	-
4	T006	J01	0	625	242	147	submit	-
4	T007	J01	0	625	242	148	submit	-
4	T008	J01	0	625	242	148	submit	-
4	T009	J01	0	625	242	148	submit	-
4	T010	J01	0	625	242	148	submit	-
4	T011	J01	0	625	242	149	submit	-
4	T012	J01	0	625	242	147	submit	-
4	T013	J01	0	625	242	148	submit	-
4	T014	J01	0	625	242	148	submit	-
4	T015	J01	0	625	242	149	submit	-
4	T016	J01	0	625	242	148	submit	-
4	T001	J01	0	625	242	149	start	M01
4	T002	J01	0	625	242	149	start	M01
4	T003	J01	0	625	242	148	start	M01
4	T004	J01	0	625	242	147	start	M01
4	T005	J01	0	625	242	149	start	M01
4	T006	J01	0	625	242	147	start	M01
4	T007	J01	0	625	242	148	start	M01
4	T008	J01	0	625	242	148	start	M01
4	T009	J01	0	625	242	148	start	M01
4	T010	J01	0	625	242	148	start	M01
4	T011	J01	0	625	242	149	start	M02
4	T012	J01	0	625	242	147	start	M02
4	T013	J01	0	625	242	148	start	M02
4	T014	J01	0	625	242	148	start	M02
4	T015	J01	0	625	242	149	start	M02
4	T016	J01	0	625	242	148	start	M02
18	T017	J02	1	3125	622	90	submit	-
18	T017	J02	1	3125	622	90	start	M03
75	T018	J03	9	4688	6360	482	submit	-
75	T018	J03	9	4688	6360	482	start	M06
108	T017	J02	1	3125	622	90	finish	M03
151	T004	J01	0	625	242	147	finish	M01

time	task	job	priority	CPU	memory	run time	event	machine
151	T006	J01	0	625	242	147	finish	M01
151	T012	J01	0	625	242	147	finish	M02
152	T003	J01	0	625	242	148	finish	M01
152	T007	J01	0	625	242	148	finish	M01
152	T008	J01	0	625	242	148	finish	M01
152	T009	J01	0	625	242	148	finish	M01
152	T010	J01	0	625	242	148	finish	M01
152	T013	J01	0	625	242	148	finish	M02
152	T014	J01	0	625	242	148	finish	M02
152	T016	J01	0	625	242	148	finish	M02
153	T001	J01	0	625	242	149	finish	M01
153	T002	J01	0	625	242	149	finish	M01
153	T005	J01	0	625	242	149	finish	M01
153	T011	J01	0	625	242	149	finish	M02
153	T015	J01	0	625	242	149	finish	M02
169	T019	J04	8	625	155	201	submit	-
169	T019	J04	8	625	155	201	start	M01
171	T020	J05	1	1353	777	2052	submit	-
171	T020	J05	1	1353	777	2052	start	M01
173	T021	J06	0	1250	777	1211	submit	-
173	T022	J06	0	1250	777	2050	submit	-
173	T023	J06	0	1250	777	1113	submit	-
173	T021	J06	0	1250	777	1211	start	M01
173	T022	J06	0	1250	777	2050	start	M01
173	T023	J06	0	1250	777	1113	start	M02
175	T024	J07	0	937	274	100	submit	-
175	T025	J07	0	937	274	159	submit	-
175	T026	J07	0	937	274	93	submit	-
175	T027	J07	0	937	274	92	submit	-
175	T028	J07	0	937	274	133	submit	-
175	T029	J07	0	937	274	87	submit	-
175	T030	J07	0	937	274	100	submit	-
175	T031	J07	0	937	274	110	submit	-
175	T032	J07	0	937	274	111	submit	-
175	T033	J07	0	937	274	93	submit	-
175	T034	J07	0	937	274	144	submit	-
175	T035	J07	0	937	274	102	submit	-
175	T036	J07	0	937	274	121	submit	-
175	T037	J07	0	937	274	128	submit	-
175	T038	J07	0	937	274	100	submit	-
175	T039	J07	0	937	274	100	submit	-
175	T024	J07	0	937	274	100	start	M01
175	T025	J07	0	937	274	159	start	M02
175	T026	J07	0	937	274	93	start	M02
175	T027	J07	0	937	274	92	start	M02
175	T028	J07	0	937	274	133	start	M02
175	T029	J07	0	937	274	87	start	M02
175	T030	J07	0	937	274	100	start	M03
175	T031	J07	0	937	274	110	start	M03
175	T032	J07	0	937	274	111	start	M03
175	T033	J07	0	937	274	93	start	M03
175	T034	J07	0	937	274	144	start	M03
175	T035	J07	0	937	274	102	start	M03
175	T036	J07	0	937	274	121	start	M04
175	T037	J07	0	937	274	128	start	M04
175	T038	J07	0	937	274	100	start	M04
175	T039	J07	0	937	274	100	start	M04

time	task	job	priority	CPU	memory	run time	event	machine
196	T040	J08	0	6250	1244	36	submit	-
196	T040	J08	0	6250	1244	36	start	M05
197	T041	J09	0	6250	1244	59	submit	-
197	T041	J09	0	6250	1244	59	start	M07
210	T042	J10	8	625	1553	518	submit	-
210	T042	J10	8	625	1553	518	start	M04
232	T040	J08	0	6250	1244	36	finish	M05
234	T043	J11	1	1250	777	1135	submit	-
234	T043	J11	1	1250	777	1135	start	M04
237	T044	J12	8	937	1553	64	submit	-
237	T044	J12	8	937	1553	64	start	M05
251	T045	J13	8	3125	466	248	submit	-
251	T045	J13	8	3125	466	248	start	M05
252	T046	J14	4	656	197	247	submit	-
252	T047	J14	4	656	197	241	submit	-
252	T048	J14	4	656	197	240	submit	-
252	T049	J14	4	656	197	237	submit	-
252	T046	J14	4	656	197	247	start	M01
252	T047	J14	4	656	197	241	start	M05
252	T048	J14	4	656	197	240	start	M05
252	T049	J14	4	656	197	237	start	M05
256	T041	J09	0	6250	1244	59	finish	M07
256	T050	J15	0	1250	16	18	submit	-
256	T050	J15	0	1250	16	18	start	M06
262	T029	J07	0	937	274	87	finish	M02
265	T051	J16	8	3125	622	24	submit	-
265	T051	J16	8	3125	622	24	start	M07
267	T027	J07	0	937	274	92	finish	M02
268	T026	J07	0	937	274	93	finish	M02
268	T033	J07	0	937	274	93	finish	M03
274	T050	J15	0	1250	16	18	finish	M06
275	T024	J07	0	937	274	100	finish	M01
275	T030	J07	0	937	274	100	finish	M03
275	T038	J07	0	937	274	100	finish	M04
275	T039	J07	0	937	274	100	finish	M04
277	T035	J07	0	937	274	102	finish	M03
277	T052	J17	8	6250	466	438	submit	-
277	T052	J17	8	6250	466	438	start	M08
285	T031	J07	0	937	274	110	finish	M03
286	T032	J07	0	937	274	111	finish	M03
289	T051	J16	8	3125	622	24	finish	M07
296	T036	J07	0	937	274	121	finish	M04
301	T044	J12	8	937	1553	64	finish	M05
303	T037	J07	0	937	274	128	finish	M04
308	T028	J07	0	937	274	133	finish	M02
319	T034	J07	0	937	274	144	finish	M03
334	T025	J07	0	937	274	159	finish	M02
361	T053	J18	8	1250	2863	1440	submit	-
361	T053	J18	8	1250	2863	1440	start	M02
370	T019	J04	8	625	155	201	finish	M01
384	T054	J19	8	625	155	185	submit	-
384	T054	J19	8	625	155	185	start	M01
386	T055	J20	1	1353	777	1093	submit	-
386	T055	J20	1	1353	777	1093	start	M02
388	T056	J21	0	1250	777	836	submit	-
388	T057	J21	0	1250	777	779	submit	-
388	T056	J21	0	1250	777	836	start	M02

time	task	job	priority	CPU	memory	run time	event	machine
388	T057	J21	0	1250	777	779	start	M03
394	T018	J03	9	4688	6360	482	fail	M06
418	T058	J22	8	3125	795	277	submit	-
418	T058	J22	8	3125	795	277	start	M03
432	T059	J23	8	937	466	81	submit	-
432	T059	J23	8	937	466	81	start	M02
489	T049	J14	4	656	197	237	finish	M05
492	T048	J14	4	656	197	240	finish	M05
493	T047	J14	4	656	197	241	finish	M05
494	T060	J24	0	3125	466	259	submit	-
494	T060	J24	0	3125	466	259	start	M04
496	T061	J25	1	3125	3180	192	submit	-
496	T061	J25	1	3125	3180	192	start	M05
499	T045	J13	8	3125	466	248	finish	M05
499	T046	J14	4	656	197	247	finish	M01
513	T059	J23	8	937	466	81	finish	M02
546	T062	J26	4	3125	472	193	submit	-
546	T063	J26	4	3125	472	225	submit	-
546	T064	J26	4	3125	472	233	submit	-
546	T065	J26	4	3125	472	203	submit	-
546	T062	J26	4	3125	472	193	start	M05
546	T063	J26	4	3125	472	225	start	M07
546	T064	J26	4	3125	472	233	start	M07
554	T066	J27	0	6250	466	38	submit	-
559	T067	J28	1	1572	3180	118	submit	-
559	T068	J29	9	6250	1590	69	submit	-
559	T069	J30	4	625	155	24	submit	-
559	T069	J30	4	625	155	24	start	M01
562	T070	J31	0	937	966	80	submit	-
562	T071	J31	0	937	966	81	submit	-
562	T072	J31	0	937	966	93	submit	-
562	T073	J31	0	937	966	110	submit	-
562	T074	J31	0	937	966	79	submit	-
562	T075	J31	0	937	966	80	submit	-
562	T076	J31	0	937	966	80	submit	-
562	T077	J31	0	937	966	78	submit	-
562	T078	J31	0	937	966	93	submit	-
562	T079	J31	0	937	966	111	submit	-
562	T080	J31	0	937	966	110	submit	-
562	T081	J31	0	937	966	95	submit	-
562	T082	J31	0	937	966	94	submit	-
562	T083	J31	0	937	966	92	submit	-
562	T084	J31	0	937	966	79	submit	-
562	T085	J31	0	937	966	81	submit	-
562	T070	J31	0	937	966	80	start	M02
562	T071	J31	0	937	966	81	start	M03
562	T072	J31	0	937	966	93	start	M04
569	T054	J19	8	625	155	185	finish	M01
583	T069	J30	4	625	155	24	finish	M01
583	T022	J06	0	1250	777	2050	evict	M01
583	T021	J06	0	1250	777	1211	evict	M01
583	T020	J05	1	1353	777	2052	evict	M01
583	T068	J29	9	6250	1590	69	start	M01
613	T086	J32	8	6250	3180	756	submit	-
642	T070	J31	0	937	966	80	finish	M02
642	T073	J31	0	937	966	110	start	M02
643	T071	J31	0	937	966	81	finish	M03

time	task	job	priority	CPU	memory	run time	event	machine
643	T020	J05	1	1353	777	2052	start	M03
652	T068	J29	9	6250	1590	69	finish	M01
652	T065	J26	4	3125	472	203	start	M01
652	T021	J06	0	1250	777	1211	start	M01
652	T022	J06	0	1250	777	2050	start	M01
655	T072	J31	0	937	966	93	finish	M04
655	T074	J31	0	937	966	79	start	M04
675	T087	J33	0	6250	1244	63	submit	-
676	T088	J34	9	4688	6360	496	submit	-
678	T089	J35	4	2725	2704	454	submit	-
678	T090	J35	4	2725	2704	441	submit	-
678	T091	J35	4	2725	2704	440	submit	-
678	T092	J35	4	2725	2704	443	submit	-
688	T061	J25	1	3125	3180	192	finish	M05
695	T058	J22	8	3125	795	277	finish	M03
695	T093	J36	8	1250	932	374	submit	-
695	T094	J37	4	312	6	23	submit	-
695	T093	J36	8	1250	932	374	start	M03
695	T094	J37	4	312	6	23	start	M01
715	T052	J17	8	6250	466	438	finish	M08
715	T089	J35	4	2725	2704	454	start	M08
718	T094	J37	4	312	6	23	finish	M01
721	T095	J38	4	937	213	48	submit	-
721	T096	J38	4	937	213	48	submit	-
721	T097	J38	4	937	213	45	submit	-
721	T098	J38	4	937	213	49	submit	-
721	T095	J38	4	937	213	48	start	M03
721	T096	J38	4	937	213	48	start	M03
721	T097	J38	4	937	213	45	start	M08
728	T042	J10	8	625	1553	518	finish	M04
728	T098	J38	4	937	213	49	start	M04
733	T099	J39	8	1250	777	52	submit	-
734	T074	J31	0	937	966	79	finish	M04
734	T075	J31	0	937	966	80	start	M04
736	T100	J40	1	6250	6360	1244	submit	-
738	T101	J41	1	6250	6360	913	submit	-
739	T062	J26	4	3125	472	193	finish	M05
739	T086	J32	8	6250	3180	756	start	M05
747	T102	J42	4	3125	1125	23	submit	-
752	T073	J31	0	937	966	110	finish	M02
752	T076	J31	0	937	966	80	start	M02
753	T060	J24	0	3125	466	259	finish	M04
753	T099	J39	8	1250	777	52	start	M04
766	T097	J38	4	937	213	45	finish	M08
769	T095	J38	4	937	213	48	finish	M03
769	T096	J38	4	937	213	48	finish	M03
771	T063	J26	4	3125	472	225	finish	M07
771	T102	J42	4	3125	1125	23	start	M07
777	T098	J38	4	937	213	49	finish	M04
777	T090	J35	4	2725	2704	441	start	M04
779	T064	J26	4	3125	472	233	finish	M07
794	T102	J42	4	3125	1125	23	finish	M07
794	T091	J35	4	2725	2704	440	start	M07
796	T103	J43	8	3125	1590	259	submit	-
797	T104	J44	4	3125	488	223	submit	-
797	T105	J44	4	3125	488	242	submit	-
797	T106	J44	4	3125	488	257	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
797	T107	J44	4	3125	488	228	submit	-
805	T099	J39	8	1250	777	52	finish	M04
805	T077	J31	0	937	966	78	start	M04
811	T108	J45	4	3125	1125	33	submit	-
814	T075	J31	0	937	966	80	finish	M04
814	T078	J31	0	937	966	93	start	M04
822	T109	J46	8	625	155	103	submit	-
822	T109	J46	8	625	155	103	start	M01
832	T076	J31	0	937	966	80	finish	M02
832	T079	J31	0	937	966	111	start	M02
854	T110	J47	9	6250	233	113	submit	-
855	T065	J26	4	3125	472	203	finish	M01
855	T104	J44	4	3125	488	223	start	M01
873	T111	J48	4	3125	1125	31	submit	-
883	T077	J31	0	937	966	78	finish	M04
883	T080	J31	0	937	966	110	start	M04
907	T078	J31	0	937	966	93	finish	M04
907	T081	J31	0	937	966	95	start	M04
925	T109	J46	8	625	155	103	finish	M01
943	T079	J31	0	937	966	111	finish	M02
943	T082	J31	0	937	966	94	start	M02
968	T112	J49	1	6250	466	455	submit	-
970	T113	J50	0	705	204	453	submit	-
970	T114	J50	0	705	204	450	submit	-
970	T115	J50	0	705	204	453	submit	-
970	T116	J50	0	705	204	451	submit	-
970	T117	J50	0	705	204	451	submit	-
970	T118	J50	0	705	204	451	submit	-
970	T113	J50	0	705	204	453	start	M03
970	T114	J50	0	705	204	450	start	M03
970	T115	J50	0	705	204	453	start	M03
970	T116	J50	0	705	204	451	start	M07
970	T117	J50	0	705	204	451	start	M07
993	T080	J31	0	937	966	110	finish	M04
993	T083	J31	0	937	966	92	start	M04
1002	T081	J31	0	937	966	95	finish	M04
1002	T084	J31	0	937	966	79	start	M04
1037	T082	J31	0	937	966	94	finish	M02
1037	T119	J51	1	6250	6360	3220	submit	-
1037	T085	J31	0	937	966	81	start	M02
1069	T093	J36	8	1250	932	374	finish	M03
1069	T115	J50	0	705	204	453	evict	M03
1069	T114	J50	0	705	204	450	evict	M03
1069	T113	J50	0	705	204	453	evict	M03
1069	T057	J21	0	1250	777	779	evict	M03
1069	T020	J05	1	1353	777	2052	evict	M03
1069	T110	J47	9	6250	233	113	start	M03
1069	T113	J50	0	705	204	453	start	M08
1069	T114	J50	0	705	204	450	start	M08
1078	T104	J44	4	3125	488	223	finish	M01
1078	T105	J44	4	3125	488	242	start	M01
1081	T084	J31	0	937	966	79	finish	M04
1081	T057	J21	0	1250	777	779	start	M04
1085	T083	J31	0	937	966	92	finish	M04
1085	T115	J50	0	705	204	453	start	M04
1118	T085	J31	0	937	966	81	finish	M02
1118	T118	J50	0	705	204	451	start	M02

time	task	job	priority	CPU	memory	run time	event	machine
1143	T120	J52	1	1250	777	1029	submit	-
1155	T121	J53	0	6250	466	47	submit	-
1156	T122	J54	0	6250	1244	47	submit	-
1160	T123	J55	1	1250	1590	386	submit	-
1169	T089	J35	4	2725	2704	454	finish	M08
1169	T103	J43	8	3125	1590	259	start	M08
1169	T020	J05	1	1353	777	2052	start	M08
1170	T124	J56	4	625	155	24	submit	-
1170	T125	J57	4	625	311	212	submit	-
1170	T126	J58	8	1250	398	24	submit	-
1170	T124	J56	4	625	155	24	start	M01
1173	T127	J59	9	6250	1590	77	submit	-
1182	T110	J47	9	6250	233	113	finish	M03
1182	T127	J59	9	6250	1590	77	start	M03
1186	T128	J60	6	625	187	227	submit	-
1186	T129	J61	6	625	280	40	submit	-
1188	T130	J62	1	1250	466	171	submit	-
1191	T131	J63	0	625	229	169	submit	-
1191	T132	J63	0	625	229	168	submit	-
1191	T133	J63	0	625	229	168	submit	-
1191	T134	J63	0	625	229	168	submit	-
1191	T135	J63	0	625	229	168	submit	-
1191	T136	J63	0	625	229	168	submit	-
1191	T137	J63	0	625	229	169	submit	-
1191	T138	J63	0	625	229	167	submit	-
1191	T139	J63	0	625	229	168	submit	-
1191	T140	J63	0	625	229	166	submit	-
1191	T141	J63	0	625	229	167	submit	-
1191	T142	J63	0	625	229	168	submit	-
1191	T143	J63	0	625	229	168	submit	-
1191	T144	J63	0	625	229	167	submit	-
1191	T145	J63	0	625	229	168	submit	-
1191	T146	J63	0	625	229	168	submit	-
1194	T124	J56	4	625	155	24	finish	M01
1194	T128	J60	6	625	187	227	start	M01

Cell window 29: Events

Time 394: machine M06 failed; task T018 stopped and went back to the queue.

Time 583: task T068 (priority 9) evicted task T020 (priority 1) from machine M01.

Time 583: task T068 (priority 9) evicted task T021 (priority 0) from machine M01.

Time 583: task T068 (priority 9) evicted task T022 (priority 0) from machine M01.

Time 643: task T020 restarted on machine M03.

Time 652: task T021 restarted on machine M01.

Time 652: task T022 restarted on machine M01.

Time 1069: task T110 (priority 9) evicted task T020 (priority 1) from machine M03.

Time 1069: task T110 (priority 9) evicted task T057 (priority 0) from machine M03.

Time 1069: task T110 (priority 9) evicted task T113 (priority 0) from machine M03.

Time 1069: task T110 (priority 9) evicted task T114 (priority 0) from machine M03.

Time 1069: task T110 (priority 9) evicted task T115 (priority 0) from machine M03.

Time 1069: task T113 restarted on machine M08.

Time 1069: task T114 restarted on machine M08.

Time 1081: task T057 restarted on machine M04.

Time 1085: task T115 restarted on machine M04.

Time 1169: task T020 restarted on machine M08.

Longest waits (submission to first start, tasks started by the window end): task T085 waited 475 s (submitted at 562, first started at 1037); task T084 waited 440 s (submitted at 562, first started at 1002); task T083 waited 431 s (submitted at 562, first started at 993); task T082 waited 381 s (submitted at 562, first started at 943); task T103 waited 373 s (submitted at 796, first started at 1169).

Cell window 29: Queue at window end

State after the events at time 1200.

Tasks running at time 1200 (remaining run time = seconds still to run after time 1200):

task	job	priority	CPU	memory	machine	started	remaining
T021	J06	0	1250	777	M01	652	663
T022	J06	0	1250	777	M01	652	1502
T105	J44	4	3125	488	M01	1078	120
T128	J60	6	625	187	M01	1194	221
T023	J06	0	1250	777	M02	173	86
T053	J18	8	1250	2863	M02	361	601
T055	J20	1	1353	777	M02	386	279
T056	J21	0	1250	777	M02	388	24
T118	J50	0	705	204	M02	1118	369
T127	J59	9	6250	1590	M03	1182	59
T043	J11	1	1250	777	M04	234	169
T057	J21	0	1250	777	M04	1081	660
T090	J35	4	2725	2704	M04	777	18
T115	J50	0	705	204	M04	1085	338
T086	J32	8	6250	3180	M05	739	295
T091	J35	4	2725	2704	M07	794	34
T116	J50	0	705	204	M07	970	221
T117	J50	0	705	204	M07	970	221
T020	J05	1	1353	777	M08	1169	2021
T103	J43	8	3125	1590	M08	1169	228
T113	J50	0	705	204	M08	1069	322
T114	J50	0	705	204	M08	1069	319

Tasks waiting at time 1200, in queue order (each will run its full run time once started):

task	job	priority	CPU	memory	submitted	run time
T018	J03	9	4688	6360	75	482
T088	J34	9	4688	6360	676	496
T126	J58	8	1250	398	1170	24
T129	J61	6	625	280	1186	40
T092	J35	4	2725	2704	678	443
T106	J44	4	3125	488	797	257
T107	J44	4	3125	488	797	228
T108	J45	4	3125	1125	811	33
T111	J48	4	3125	1125	873	31
T125	J57	4	625	311	1170	212
T067	J28	1	1572	3180	559	118
T100	J40	1	6250	6360	736	1244
T101	J41	1	6250	6360	738	913
T112	J49	1	6250	466	968	455

task	job	priority	CPU	memory	submitted	run time
T119	J51	1	6250	6360	1037	3220
T120	J52	1	1250	777	1143	1029
T123	J55	1	1250	1590	1160	386
T130	J62	1	1250	466	1188	171
T066	J27	0	6250	466	554	38
T087	J33	0	6250	1244	675	63
T121	J53	0	6250	466	1155	47
T122	J54	0	6250	1244	1156	47
T131	J63	0	625	229	1191	169
T132	J63	0	625	229	1191	168
T133	J63	0	625	229	1191	168
T134	J63	0	625	229	1191	168
T135	J63	0	625	229	1191	168
T136	J63	0	625	229	1191	168
T137	J63	0	625	229	1191	169
T138	J63	0	625	229	1191	167
T139	J63	0	625	229	1191	168
T140	J63	0	625	229	1191	166
T141	J63	0	625	229	1191	167
T142	J63	0	625	229	1191	168
T143	J63	0	625	229	1191	168
T144	J63	0	625	229	1191	167
T145	J63	0	625	229	1191	168
T146	J63	0	625	229	1191	168

Machine M06 is down at time 1200; it returns at time 1347.